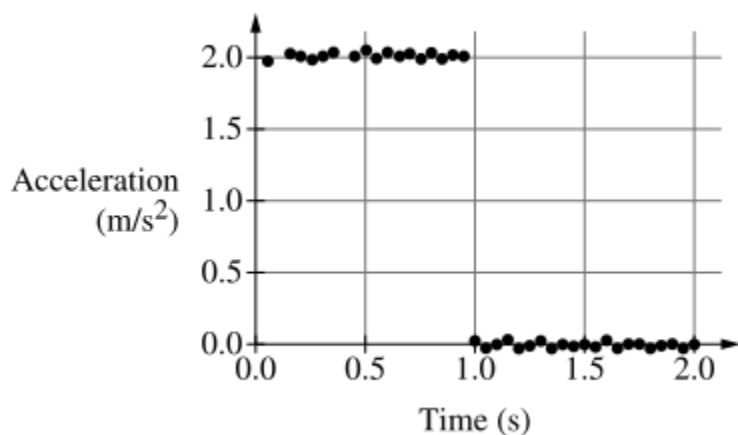


U1

1. A cart is moving along a straight track and has velocity $+0.5 \text{ m/s}$ and positive acceleration at time 1 s . At time 3 s , the cart reaches a maximum speed. At time 4 s , the cart again has velocity $+0.5 \text{ m/s}$, but with negative acceleration. A student claims that the average acceleration of the cart between time 1 s and 4 s time is zero. Is the student's claim correct? Why or why not?
- (A) Yes, because the final velocity of the cart is equal to the initial velocity.
- (B) Yes, because the cart returns to its original position.
- (C) No, because the values of the acceleration at time 1 s and time 4 s are unknown.
- (D) No, because the cart spends more time with a positive acceleration than with a negative acceleration.
2. A car is driving east on a straight, level road with a speed of 20 m/s when it begins to accelerate eastward at a constant rate. After 5 s the car has a speed of 40 m/s to the east. The acceleration of the car during the 5 s time interval is
- (A) 4 m/s^2
- (B) 8 m/s^2
- (C) 20 m/s^2
- (D) 30 m/s^2
- 3.



A cart begins to move from rest on a horizontal track. The acceleration of the cart as a function of time is shown in the graph. Which of the following correctly indicates the magnitude of the average velocity of the cart during the interval shown, and provides a valid explanation?

- (A) Less than 2 m/s because the cart has a velocity of 2 m/s for the first half of the interval and zero velocity for the second half.
- (B) Less than 2 m/s because the cart is traveling at a speed of 2 m/s or less during the time interval.
- (C) Equal to 2 m/s because the cart is initially at rest and then moves at a constant speed of 2 m/s during the second half of the interval.
- (D) Equal to 2 m/s because the cart has an average acceleration of 1 m/s^2 over an interval of 2 s .

U1

4. An object on a straight, horizontal track with an initial positive velocity undergoes a constant acceleration of -5 m/s^2 for 4 seconds. Which of the following is the change in velocity of the object during this interval?
- (A) -20 m/s
(B) -5 m/s
(C) -1.2 m/s
(D) 0 m/s

5.

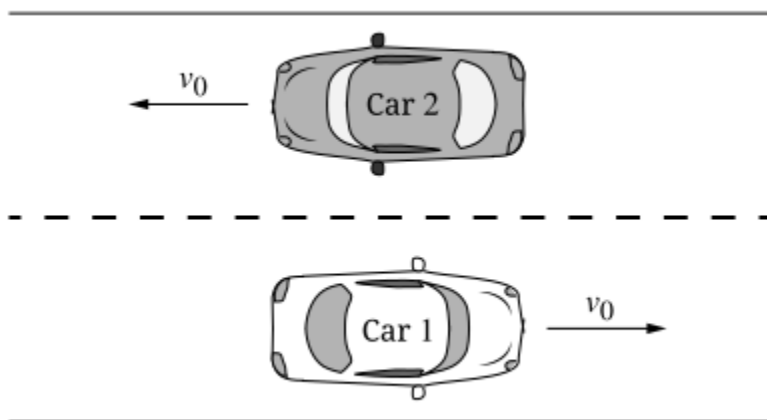
Interval name	Average Speed	Final Position
A	5.0 m/s	25 m
B	4.0 m/s	0 m
C	6.0 m/s	75 m
D	2.0 m/s	0 m
E	7.0 m/s	50 m

An athlete sprints back and forth along a straight track. The athlete starts at a position of 0 m, and the athlete's average speed and final position are shown in the table for five successive time intervals. Assume the athlete has a constant speed during each interval and quickly changes direction during the transitions between intervals. During which transition does the athlete's change in velocity have the greatest magnitude?

- (A) The transition between intervals A and B.
(B) The transition between intervals B and C.
(C) The transition between intervals C and D.
(D) The transition between intervals D and E.

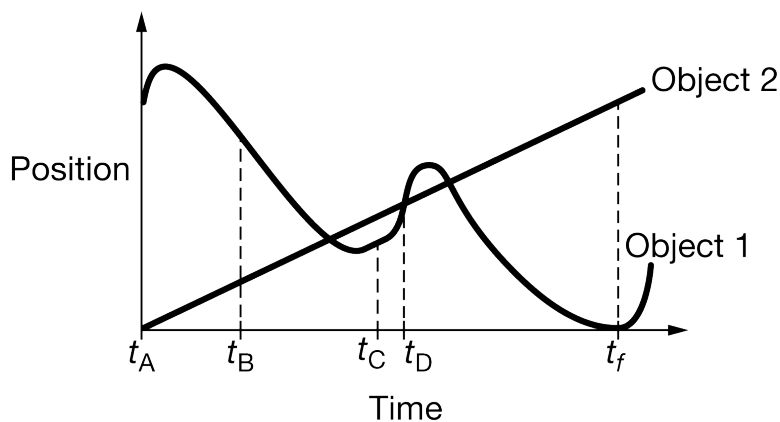
U1

6.



Two cars pass by each other on a straight road. Car 1 has a speed v_0 to the right and Car 2 has the same speed to the left. Immediately after passing each other, both cars experience a constant acceleration a_0 directed to the left. At the instant Car 1 comes to rest, what is the speed v_2 of Car 2?

- (A) $v_2 = 0$
- (B) $0 < v_2 < 2v_0$
- (C) $v_2 = 2v_0$
- (D) $v_2 > 2v_0$



The position as a function of time for two objects moving along a straight line is shown in the graph.

7. Which statement is true about the distances the two object have traveled at time t_f ?

- (A) Object 1 has traveled a greater distance.
- (B) Object 2 has traveled a greater distance.
- (C) Both objects have traveled the same distance.
- (D) The total distance traveled by each object cannot be compared using the graph.

U1

8. At which of the following times do the two objects have the same velocity?

- (A) t_A
 - (B) t_B
 - (C) t_C
 - (D) t_D
-

9.

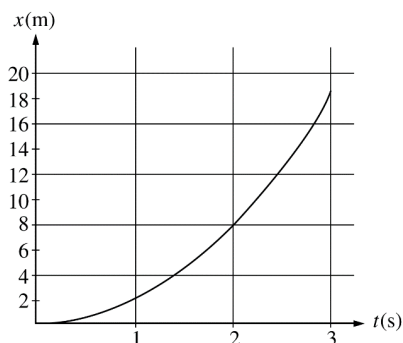
	Initial Velocity	Final Velocity
Cart 1	30 cm/s to the right	20 cm/s to the right
Cart 2	30 cm/s to the left	20 cm/s to the left
Cart 3	10 cm/s to the left	10 cm/s to the right

Three carts are free to move left or right on a horizontal track. Each cart experiences a constant acceleration. The initial and final velocities of each cart are indicated in the table. Which cart, if any, has a leftward acceleration?

- (A) Cart 1
 - (B) Cart 2
 - (C) Cart 3
 - (D) None of the carts has a leftward acceleration.
10. A 1.5 kg cart starts from rest and then accelerates at a constant rate of 2 m/s^2 to the right for 5 s. The displacement of the cart over this time interval is most nearly
- (A) 3 m
 - (B) 10 m
 - (C) 15 m
 - (D) 25 m
11. A person driving a car suddenly applies the brakes. The car takes 4 s to come to rest while traveling 20 m at constant acceleration. Can the speed of the car immediately before the brakes were applied be determined without first determining the car's acceleration?
- (A) Yes, by dividing the distance (20 m) by the time (4 s).
 - (B) Yes, by determining the average speed while braking and doubling it.
 - (C) No, because the acceleration is needed to use standard equations such as $\Delta x = v_o t + 1/2 at^2$.
 - (D) No, because the fundamental relationship that defines velocity contains acceleration.
-

U1

12.



The graph above represents position x versus time t for an object being acted on by a constant force. The average speed during the interval between 1 s and 2 s is most nearly

- (A) 2 m/s
- (B) 4 m/s
- (C) 5 m/s
- (D) 6 m/s
- (E) 8 m/s

13. For which of the following motions of an object must the acceleration always be zero?

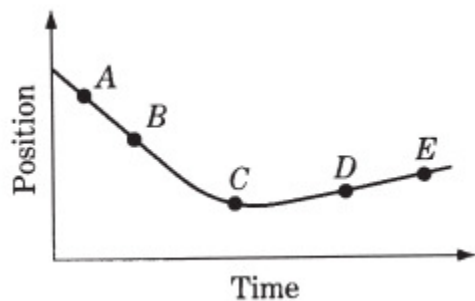
I. Any motion in a straight line

II. Simple harmonic motion

III. Any motion in a circle

- (A) I only
- (B) II only
- (C) III only
- (D) Either I or III, but not II
- (E) None of these motions guarantees zero acceleration.

14.



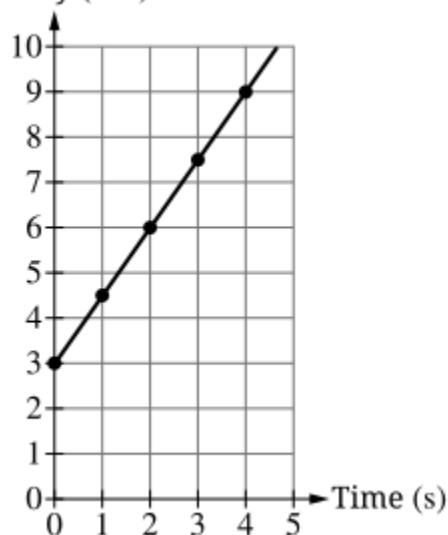
The motion of a particle along a straight line is represented by the position versus time graph above. At which of the labeled points on the graph is the magnitude of the acceleration of the particle greatest?

U1

- (A) *A*
- (B) *B*
- (C) *C*
- (D) *D*
- (E) *E*

15.

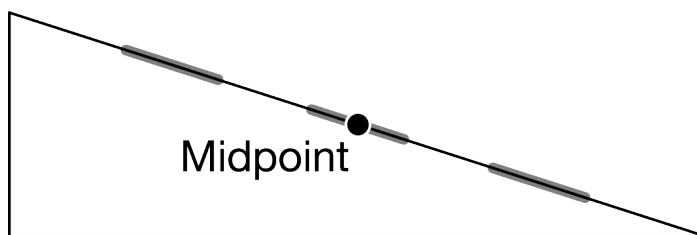
Velocity (m/s)



The velocity of an object as a function of time is shown in the graph. Which of the following is most nearly the average acceleration of the object?

- (A) 1.5 m/s^2
- (B) 3 m/s^2
- (C) 9 m/s^2
- (D) 24 m/s^2

16.

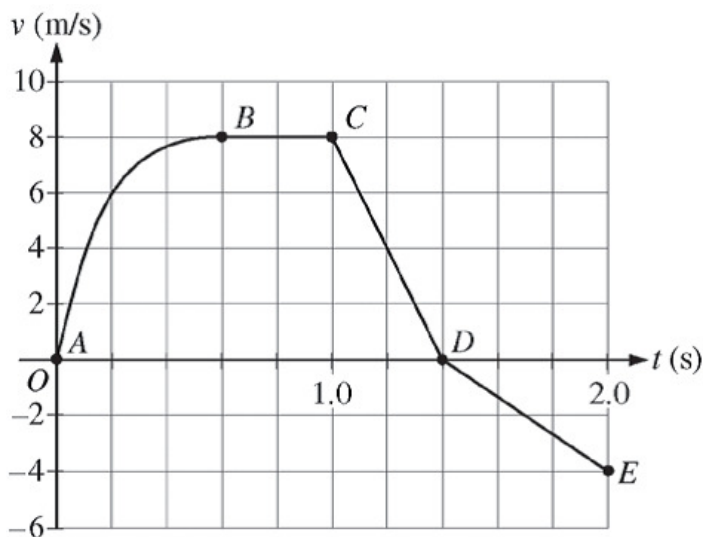


A block is released from rest and slides down a ramp. The surface of the ramp has three rough sections where the friction between the surface and the block is not negligible, as shown by the shaded regions above. Measuring which of the following will allow for the best estimate of the block's instantaneous acceleration when the block is at the midpoint of the ramp?

U1

- (A) The total distance traveled by the block and the total elapsed time
- (B) The final speed of the block and the total elapsed time
- (C) The distance between points just before and just after the midpoint and the time it takes the block to travel between them
- (D) The speed of the block at points just before and just after the midpoint and the time it takes the block to travel between them
17. A spring-loaded toy is launched straight upward from the surface of a table. In Case 1, the toy is caught at the instant that it reaches its highest point and held at rest. In Case 2, the toy is allowed to rise, then fall back to the table. In which case is the magnitude of the average velocity of the toy's free-fall motion the greatest?
- (A) Case 1, because the toy travels for less time
- (B) Case 1, because the toy has a greater displacement
- (C) Case 2, because the toy travels a greater distance
- (D) Case 2, because the toy travels for a longer time

Refer to the following material for answering the questions.



A cart is constrained to move along a straight line. A varying net force along the direction of motion is exerted on the cart. The cart's velocity v as a function of time t is shown in the graph above. The five labeled points divide the graph into four sections.

18. Which of the following correctly ranks the magnitude of the average acceleration of the cart during the four sections of the graph?
- (A) $a_{CD} > a_{AB} > a_{BC} > a_{DE}$
- (B) $a_{BC} > a_{AB} > a_{CD} > a_{DE}$
- (C) $a_{AB} > a_{BC} > a_{DE} > a_{CD}$
- (D) $a_{CD} > a_{AB} > a_{DE} > a_{BC}$

U1

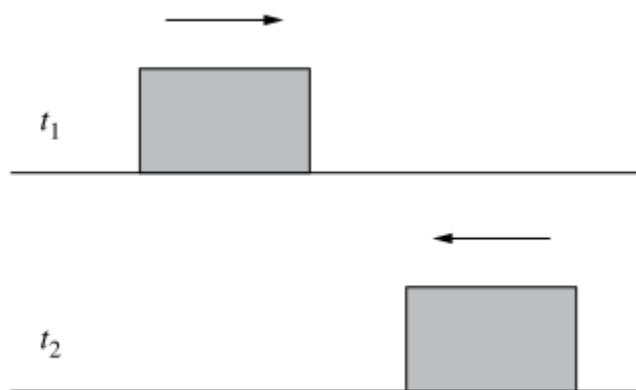
19. For which segment does the cart move the greatest distance?

- (A) AB
- (B) BC
- (C) CD
- (D) DE

20. An object is initially at the origin. The object changes its position to the right by 4 units. The object then changes its position to the left by 3 units. Which of the following correctly indicates the magnitude of the object's change in position and provides a valid justification?

- (A) 1 unit, because in one dimension vectors with opposite directions have opposite signs.
- (B) 1 unit, because the magnitude of the difference of two vectors is always equal to the difference between the magnitudes of the vectors.
- (C) 7 units, because the magnitude of the sum of two vectors is always equal to the sum of the magnitudes of the vectors.
- (D) 7 units, because the object moves a distance of 4 units followed by a distance of 3 units.

21.



A student slides a box across a level floor in a straight, horizontal path as shown in the picture. At time t_1 , the box is moving to the right at 1 m/s. At time t_2 , the box is moving to the left at 1 m/s. Which of the following correctly describes the speed and velocity at the times t_1 and t_2 ?

	Speed	Velocity
A	Different at t_1 and t_2	The same at t_1 and t_2
B	Different at t_1 and t_2	Different at t_1 and t_2
C	The same at t_1 and t_2	Different at t_1 and t_2
D	The same at t_1 and t_2	The same at t_1 and t_2

U1

- (A) A
- (B) B
- (C) C
- (D) D

22. Vectors $\vec{A}$ and $\vec{B}$ are parallel to the x -axis. Vector $\vec{C}$ is defined to be the vector sum of $\vec{A}$ and $\vec{B}$, or $\vec{C} = \vec{A} + \vec{B}$. Vector $\vec{C}$ has the same direction as vector $\vec{B}$ and the magnitudes of $\vec{B}$ and $\vec{C}$ are related by $|\vec{B}| > |\vec{C}|$. Which of the following claims is correct about the relative directions and magnitudes of vectors $\vec{A}$ and $\vec{B}$?

- (A) $|\vec{A}| > |\vec{B}|$ and the vectors have the same direction.
- (B) $|\vec{A}| > |\vec{B}|$ and the vectors have opposite directions.
- (C) $|\vec{A}| < |\vec{B}|$ and the vectors have the same direction.
- (D) $|\vec{A}| < |\vec{B}|$ and the vectors have opposite directions.

23. During time interval Δt_1 , a car starts from rest and speeds up at a constant rate to the right until reaching a final velocity v_f . Then, during time interval Δt_2 the car slows down at a constant rate until it comes to rest. Which of the following correctly indicates whether the signs and directions of the car's acceleration during the two time intervals are the same or different?

	Signs	Directions
A	Same	Same
B	Same	Different
C	Different	Same
D	Different	Different

- (A) A
- (B) B
- (C) C
- (D) D

24. Vectors $\vec{M}$ and $\vec{N}$ are parallel to the y -axis. Let $a = |\vec{M} + \vec{N}|$ and $b = |\vec{M}| + |\vec{N}|$. A student claims that is possible that $a < b$. Which of the following indicates whether the claim is correct and provides a valid justification?

U1

- (A) The claim is correct, because the magnitude of the sum of two vectors is always less than the sum of the magnitudes of the vectors.
- (B) The claim is correct, because the vectors $\vec{M}$ and $\vec{N}$ could be in opposite directions.
- (C) The claim is incorrect, because the values of a and b must be equal since the vectors $\vec{M}$ and $\vec{N}$ are parallel.
- (D) The claim is incorrect, because the values of a and b must be equal since the magnitude of a vector sum must equal the sum of the individual vector magnitudes.
25. A cart moves along a straight, level track. At time $t = 0$ s, the cart has a speed of v_0 to the right. At time $t = 1$ s, the cart has a speed of v_1 to the left. Which of the following correctly describes the magnitude and direction of the change $\Delta \vec{v}$ in the cart's velocity between $t = 0$ s and $t = 1$ s?
- (A) $v_1 + v_0$ to the right
- (B) $v_1 + v_0$ to the left
- (C) $|v_1 - v_0|$ to the right
- (D) $|v_1 - v_0|$ to the left

26.

Car	Velocity	Acceleration
A	+6.0 m/s	+1.5 m/s ²
B	−7.0 m/s	+2.0 m/s ²
C	+8.0 m/s	−2.5 m/s ²
D	−9.0 m/s	−1.0 m/s ²

Cars A, B, C, and D are moving along a straight track. The table indicates their velocities and accelerations at the same instant. Which car has the acceleration with the greatest magnitude?

- (A) Car A
- (B) Car B
- (C) Car C
- (D) Car D

U1

27.

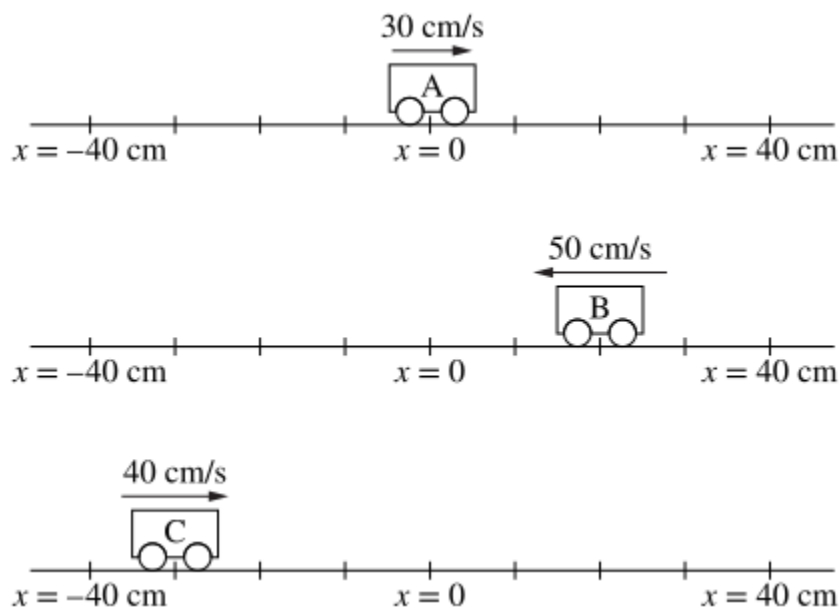


Figure A and Figure B both show two velocity vectors. Let $\vec{A}$ and $\vec{B}$ be the vector sums of the velocity vectors in Figure A and Figure B, respectively. Are the speed and the velocity of $\vec{A}$ the same as the speed and velocity of $\vec{B}$?

- (A) Yes, because both vectors have the same magnitude.
- (B) Yes, because both vectors point in the same direction.
- (C) No, because $\vec{A}$ has a magnitude of $+5 \text{ m/s}$ and $\vec{B}$ has a magnitude of -5 m/s .
- (D) No, because the vectors are in opposite directions, which means the velocities are different.

U1

28.



Three carts move along the x -axis. The locations and instantaneous velocities of the carts are shown in the figure. Which of the following correctly ranks the carts according to the magnitude of their position vectors relative to the origin at the instant shown?

- (A) $B > C > A$
- (B) $C > B > A$
- (C) $B > A > C$
- (D) $C > A > B$

29.

Position (m)	Time (s)
0.0	0.0
0.1	1.0
0.4	2.0
0.9	3.0
1.6	4.0

A block of mass 2.0 kg, starting from rest, is pushed with a constant force across a horizontal track. The position of the block as a function of time is recorded, and the data are shown in the table. What is the magnitude of the change in momentum of the block between zero and 4.0 seconds?

- (A) 0.8 kg·m/s
- (B) 1.2 kg·m/s
- (C) 1.6 kg·m/s
- (D) 3.2 kg·m/s

U1

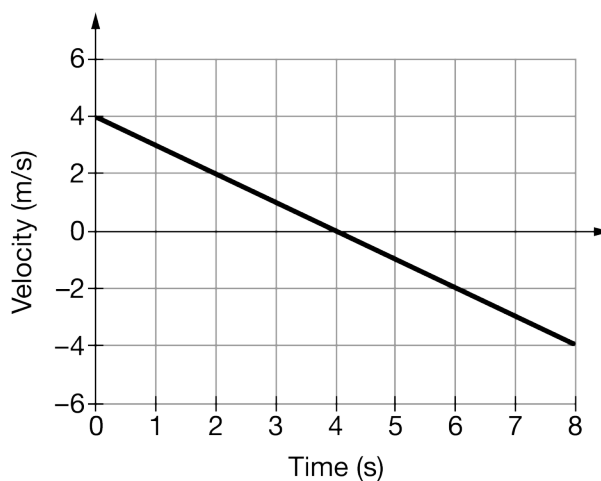
An object is thrown with an initial speed v near the surface of Earth. Assume that air resistance is negligible and the gravitational field is constant.

30. If the object is thrown vertically upward, the direction and magnitude of its acceleration while it is in the air is
- (A) upward and decreasing
 - (B) upward and constant
 - (C) downward and decreasing
 - (D) downward and increasing
 - (E) downward and constant
31. An object is thrown with an initial speed u near the surface of Earth. Assume that air resistance is negligible and the gravitational field is constant.

If the object is thrown horizontally, the direction and magnitude of its acceleration while it is in the air is

- (A) upward and decreasing
- (B) upward and constant
- (C) downward and decreasing
- (D) downward and increasing
- (E) downward and constant

32.

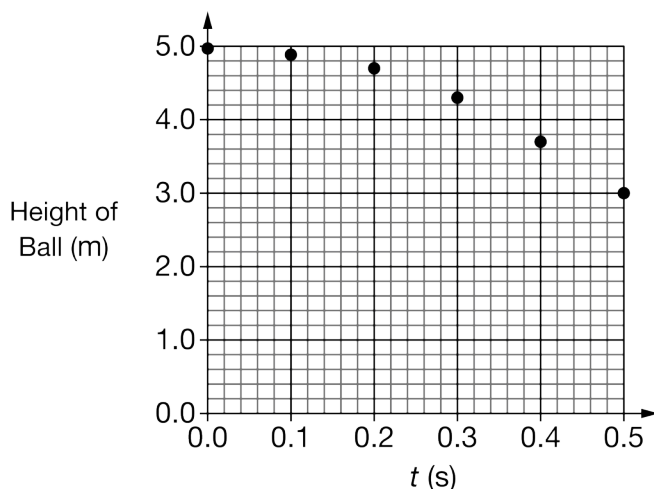


An object is sliding to the right along a straight line on a horizontal surface. The graph shows the object's velocity as a function of time. What is the object's displacement during the time depicted in the graph?

- (A) 0 m
- (B) 1 m
- (C) 8 m
- (D) 16 m

U1

33.

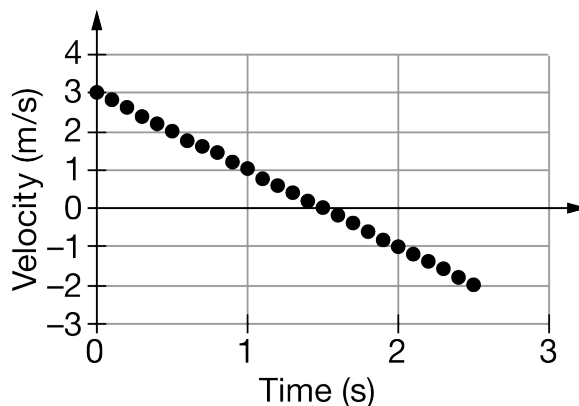


On another planet, a ball is in free fall after being released from rest at time $t = 0$. A graph of the height of the ball above the planet's surface as a function of time t is shown. The acceleration due to gravity on the planet is most nearly

- (A) 8 m/s^2
 - (B) 16 m/s^2
 - (C) 20 m/s^2
 - (D) 40 m/s^2
34. At time $t = 0$, a cart is at $x = 10 \text{ m}$ and has a velocity of 3 m/s in the $-x$ -direction. The cart has a constant acceleration in the $+x$ -direction with magnitude $3 \text{ m/s}^2 < a < 6 \text{ m/s}^2$. Which of the following gives the possible range of the position of the cart at $t = 1 \text{ s}$?
- (A) $4 \text{ m} < x < 5.5 \text{ m}$
 - (B) $8.5 \text{ m} < x < 10 \text{ m}$
 - (C) $11.5 \text{ m} < x < 13 \text{ m}$
 - (D) $14.5 \text{ m} < x < 16 \text{ m}$
35. A ball is released from rest from the twentieth floor of a building. After 1 s , the ball has fallen one floor such that it is directly outside the nineteenth-floor window. The floors are evenly spaced. Assume air resistance is negligible. What is the number of floors the ball would fall in 3 s after it is released from the twentieth floor?
- (A) 3 floors or less
 - (B) 4 to 6 floors
 - (C) 7 to 10 floors
 - (D) 11 floors or more

U1

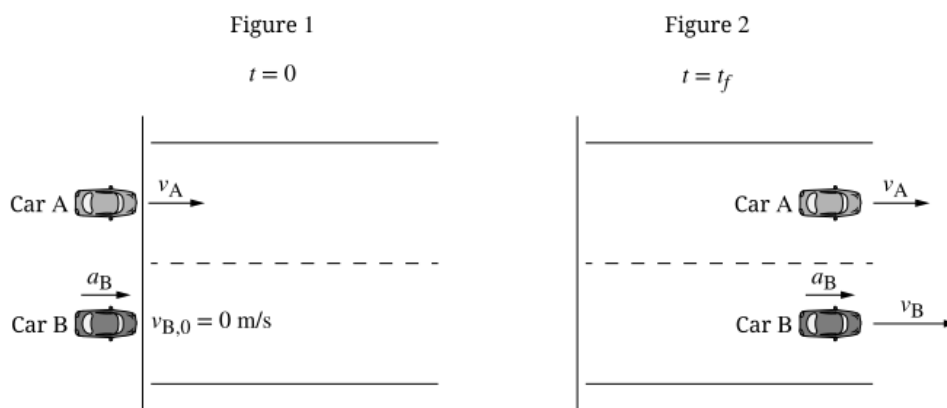
36.



At time $t = 0$, a moving cart on a horizontal track is at position 0.5 m. Using a motion detector, students generate a graph of the cart's velocity as a function of time, as shown above. At $t = 2.5$ s, the cart's position is most nearly

- (A) 0.5 m
- (B) 1.25 m
- (C) 1.75 m
- (D) 2 m

37.

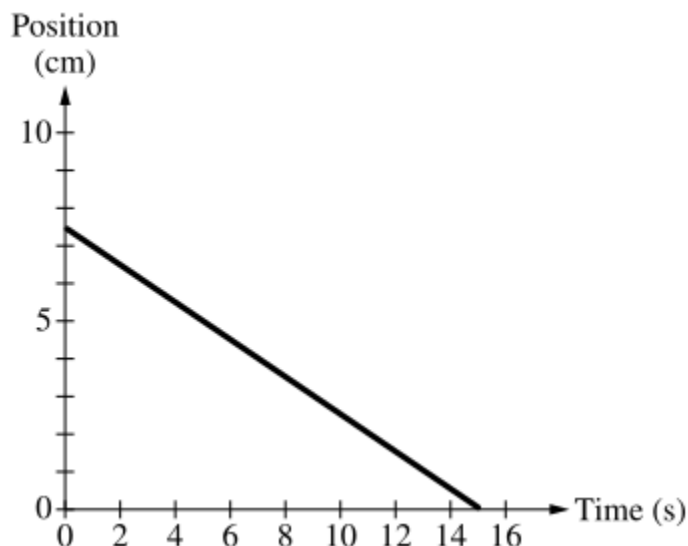


Car A is traveling to the right at a constant velocity v_A . At time $t = 0$, it passes Car B, which is at rest. At the same time ($t = 0$), Car B begins to accelerate with a constant acceleration of magnitude a_B , as shown in Figure 1. Car B has a velocity of v_B when it reaches the same position as Car A at time $t = t_f$, as shown in Figure 2. Which of the following, if any, is an expression for the time it takes for Car B to catch up to Car A?

- (A) $\frac{v_A}{a_B}$
- (B) $\frac{2v_A}{a_B}$
- (C) $\frac{2v_A^2}{a_B}$
- (D) It cannot be determined without knowing the distance traveled by the cars.

U1

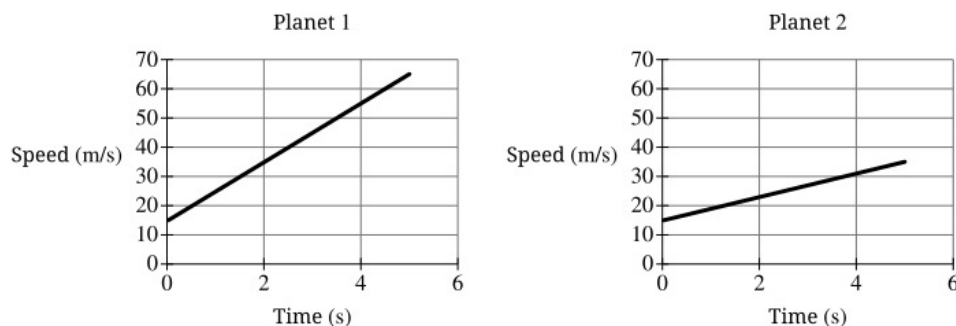
38.



A cart moves along a horizontal, one-dimensional track according to the position-time graph shown, where positive position values represent locations to the right of the origin. Which of the following statements correctly describe the cart's instantaneous velocity at time 5 s?

- (A) The cart is moving left with speed 1.0 cm/s.
- (B) The cart is moving right with speed 1.0 cm/s.
- (C) The cart is moving left with speed 0.5 cm/s.
- (D) The cart is moving right with speed 0.5 cm/s.

39.

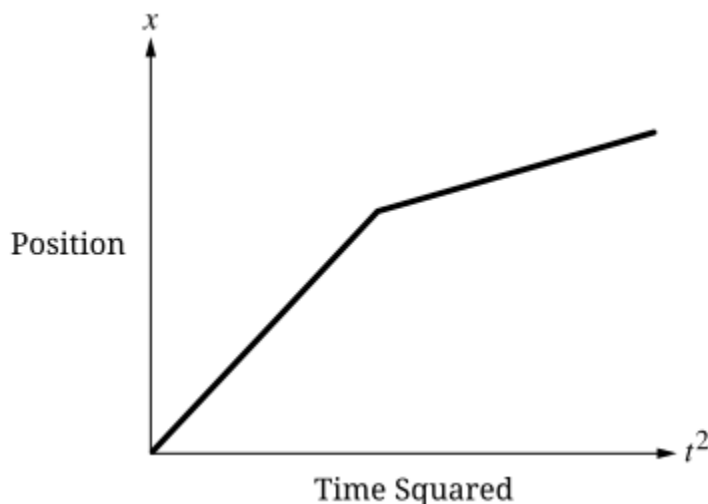


The graphs represent the speed of a ball thrown downward as a function of time near the surface of two planets. Which of the following correctly relates the acceleration a_1 of the ball near the surface of Planet 1 and the acceleration a_2 of the ball near the surface of Planet 2?

- (A) $a_1 > a_2 > 0$
- (B) $a_2 > a_1 > 0$
- (C) $(a_1 = a_2) > 0$
- (D) $a_1 = a_2 = 0$

U1

40.



A car starts from position $x = 0$ at time $t = 0$ and drives along a straight track. The graph shows the position of the car as a function of t^2 . Which of the following statements correctly describes the motion of the car?

- (A) The car moves with constant speed, momentarily stops, and then moves again with the same speed
 - (B) The car is speeding up at a constant rate, then continues to speed up but at a slower constant rate
 - (C) The car moves with constant speed, then it quickly slows down to a different, constant speed
 - (D) The car is speeding up at a constant rate, then slows down at a constant rate
41. A blue sphere and a red sphere with the same diameter are released from rest at the top of a ramp. The red sphere takes a longer time to reach the bottom of the ramp. The spheres are then rolled off a horizontal table at the same time with the same speed and fall freely to the floor. Which sphere reaches the floor first?
- (A) The red sphere
 - (B) The blue sphere
 - (C) The sphere with the greater mass
 - (D) Neither; the spheres reach the floor at the same time.
42. A car goes from rest to 30 m/s in 12 s with constant acceleration. How long does it take the car to go from rest to 15 m/s with the same acceleration?
- (A) 3.0s
 - (B) $12/\sqrt{2}$ s
 - (C) 6.0s
 - (D) $12\sqrt{2}$ s
 - (E) 24s
43. Balls 1 and 2 are each thrown horizontally from the same height above level ground, but ball 2 has a greater initial velocity after leaving the thrower's hand. If air resistance is negligible, how do the accelerations of the balls and the times it takes them to hit the ground compare?

U1

(A)	Acceleration a	Time to Hit Ground
	Greater for ball 2	Greater for ball 2

(B)	Acceleration a	Time to Hit Ground
	Greater for ball 2	Equal

(C)	Acceleration a	Time to Hit Ground
	Equal	Greater for ball 2

(D)	Acceleration a	Time to Hit Ground
	Equal	Less for ball 2

(E)	Acceleration a	Time to Hit Ground
	Equal	Equal

U1

EXPERIMENT 1
BALL RELEASED
FROM REST

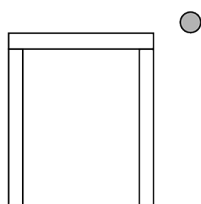


Figure 1

EXPERIMENT 2
BALL HORIZONTALLY
ROLLS OFF TABLE

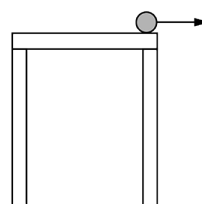
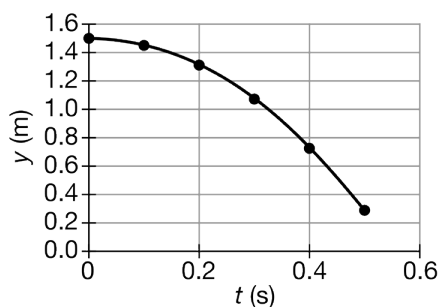
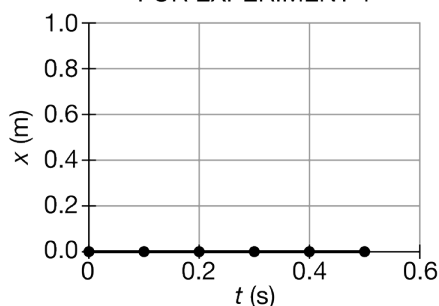
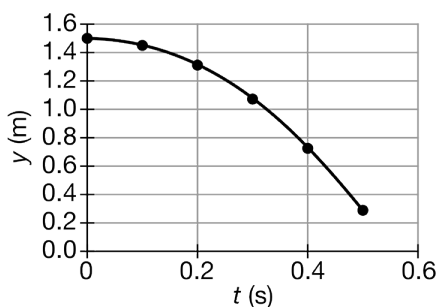
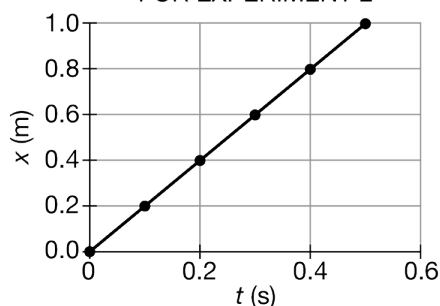


Figure 2

POSITION VS. TIME GRAPHS
FOR EXPERIMENT 1



POSITION VS. TIME GRAPHS
FOR EXPERIMENT 2

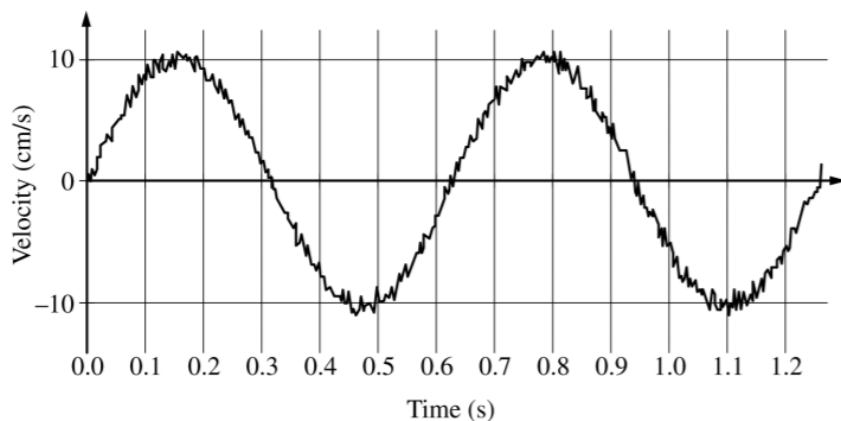


A student wants to investigate the motion of a ball by conducting two different experiments, as shown in Figure 1 and Figure 2 above. In Experiment 1, the student releases a ball from rest and uses a slow-motion camera to film the ball as it falls to the ground. Using video analysis, the student is able to plot the ball's horizontal position x and vertical position y as a function of time t . In Experiment 2, the student horizontally rolls the same ball off a table, and uses video analysis to plot the ball's horizontal position x and vertical position y as a function of time t starting from the instant the ball leaves the table. The graphs from each experiment are shown above along with each graph's best-fit curve line.

44. Suppose that Experiment 1 and Experiment 2 are conducted at the same time; one student drops the ball from rest at the same instant that a second student horizontally rolls an identical off the table. After both balls have traveled half their vertical distance to the floor, what is the acceleration of the center of mass of the two-ball system relative to Earth?

U1

- (A) Greater than g
(B) Equal to g
(C) Less than g , but not zero
(D) Zero
45. Which of the following conclusions can the student draw from the graphs, and why?
- (A) The horizontally rolled ball travels a greater horizontal distance and takes a longer amount of time to reach the ground.
(B) The released ball travels with a slower speed and takes a longer amount of time to reach the ground.
(C) Since both balls have zero horizontal acceleration, they reach the ground at the same time.
(D) Since the balls have the same vertical position at any given time, they reach the ground at the same time.
46. In Experiment 1, what is the speed of the ball the instant it makes contact with the ground?
- (A) $0 \frac{\text{m}}{\text{s}}$
(B) $0.55 \frac{\text{m}}{\text{s}}$
(C) $1.5 \frac{\text{m}}{\text{s}}$
(D) $5.4 \frac{\text{m}}{\text{s}}$
-



A student sets an object attached to a spring into oscillatory motion and uses a motion detector to record the velocity of the object as a function of time. A portion of the recorded data is shown in the figure above.

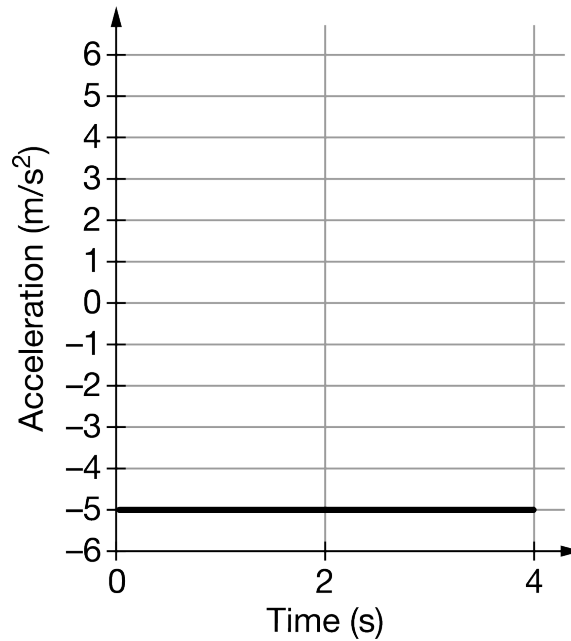
47. The acceleration of the object at time $t = 0.7 \text{ s}$ is most nearly equal to which of the following?
- (A) The value of the graph where it crosses the 0.7 s grid line
(B) The slope of the line connecting the origin and the point where the graph crosses the 0.7 s grid line
(C) The area under the curve between where the graph crosses the time axis near 0.63 s and time 0.7 s
(D) The slope of the tangent to a best-fit sinusoidal curve at 0.7 s
-

U1

48. The total change in the object's speed between 1.0 s and 1.1 s is most nearly

- (A) zero
 - (B) 5 cm / s
 - (C) 10 cm / s
 - (D) 15 cm / s
-

49.

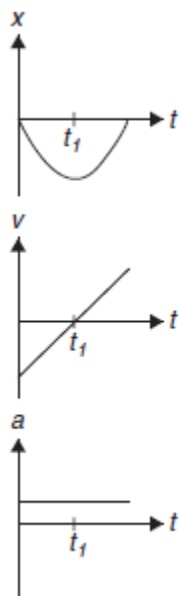


An object is released from rest near a planet's surface. A graph of the acceleration as a function of time for the object is shown for the 4 s after the object is released. The positive direction is considered to be upward. What is the displacement of the object after 2 s?

- (A) -20 m
 - (B) -10 m
 - (C) 10 m
 - (D) 20 m
-

U1

50.

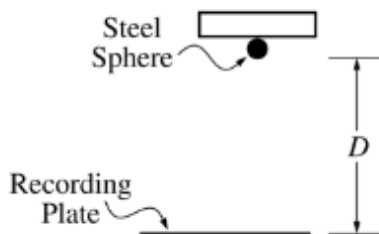


The graphs above represent the position x , velocity v , and acceleration a as a function of time t for a marble moving in one dimension. Which of the following could describe the motion of the marble?

- (A) Rolling along the floor and then bouncing off a wall
 - (B) Rolling down one side of a bowl and then rolling up the other side
 - (C) Rolling up a ramp and then rolling back down
 - (D) Falling and then bouncing elastically off a hard floor
51. A kitten sits in a lightweight basket near the edge of a table. A person accidentally knocks the basket off the table. As the kitten and basket fall, the kitten rolls, turns, kicks, and catches the basket in its claws. The basket lands on the floor with the kitten safely inside. If air resistance is negligible, what is the acceleration of the kitten-basket system while the kitten and basket are in midair?
- (A) The acceleration is directed downward with magnitude less than g because the basket is light.
 - (B) The acceleration is directed downward with magnitude equal to g because the system is a projectile.
 - (C) The acceleration fluctuates because of the rolling, turning, and kicking motion of the kitten.
 - (D) The acceleration cannot be determined without knowing how hard the basket is pushed.
52. An automobile traveling on a straight, level road has an initial speed v when the brakes are applied. In coming to rest with a constant acceleration, it travels a distance x . How far would the automobile travel in coming to rest if it had the same acceleration but an initial speed $2v$?
- (A) $\frac{1}{4}x$
 - (B) $\frac{1}{2}x$
 - (C) x
 - (D) $2x$
 - (E) $4x$

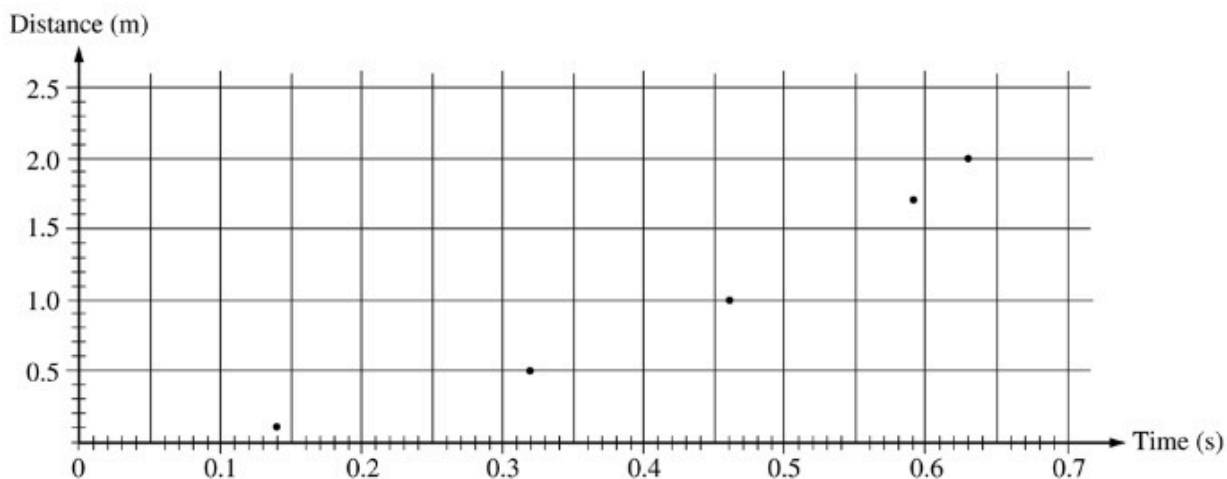
U1

53.



A student wishing to determine experimentally the acceleration g due to gravity has an apparatus that holds a small steel sphere above a recording plate, as shown above. When the sphere is released, a timer automatically begins recording the time of fall. The timer automatically stops when the sphere strikes the recording plate. The student measures the time of fall for different values of the distance D shown above and records the data in the table below. These data points are also plotted on the graph.

Distance of Fall (m)	0.10	0.50	1.00	1.70	2.00
Time of Fall (s)	0.14	0.32	0.46	0.59	0.63



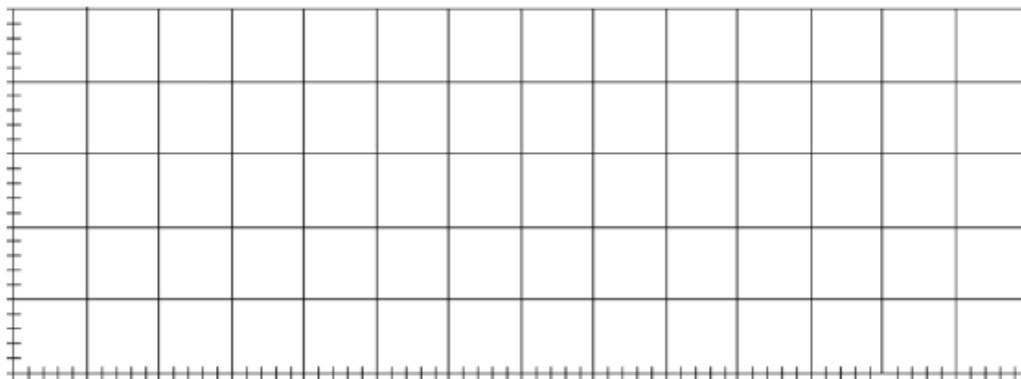
(a) On the grid above, sketch the smooth curve that best represents the student's data.

The student can use these data for distance D and time t to produce a second graph from which the acceleration g due to gravity can be determined.

(b) If only the variables D and t are used, what quantities should the student graph in order to produce a linear relationship between the two quantities?

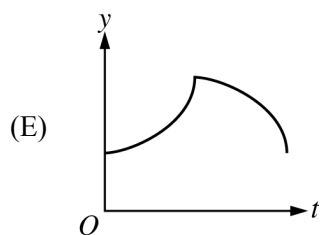
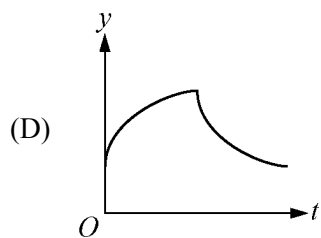
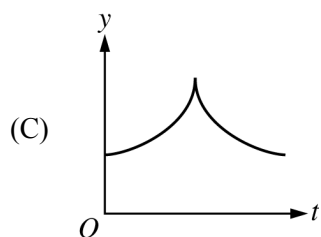
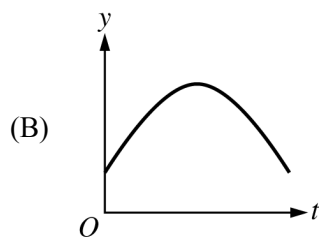
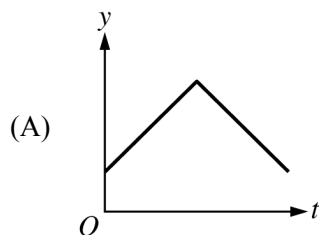
(c) On the grid below, plot the data points for the quantities you have identified in part (b), and sketch the best straight-line fit to the points. Label your axes and show the scale that you have chosen for the graph.

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- (d) Using the slope of your graph in part (c), calculate the acceleration g due to gravity in this experiment.
- (e) State one way in which the student could improve the accuracy of the results if the experiment were to be performed again. Explain why this would improve the accuracy.
54. An object is thrown with a horizontal velocity of 20 m/s from a cliff that is 125 m above level ground. If air resistance is negligible, the time that it takes the object to fall to the ground from the cliff is most nearly
- (A) 3 s
 - (B) 5 s
 - (C) 6 s
 - (D) 12 s
 - (E) 25 s
55. An object starts from rest and slides with negligible friction down an air track tipped at an angle θ from the horizontal. A student records values of the object's position along the track at various times. The value of θ can best be determined from which of the following?
- (A) The y-intercept of a graph of position as a function of time
 - (B) The y-intercept of a graph of position as a function of the square of time
 - (C) The slope of a graph of position as a function of time
 - (D) The slope of a graph of position as a function of the square of time
56. A person throws a marble straight up into the air, releasing it a short height above the ground and catching it at that same height. If air resistance is negligible, which of the following graphs of position y versus time t is correct for the motion of the marble as it goes up and then comes down?

U1



57. A small cart is rolling freely on an inclined ramp with a constant acceleration of 0.50 m/s^2 in the $-x$ direction. At time $t = 0$, the cart has a velocity of 2.0 m/s in the $+x$ -direction. If the cart never leaves the ramp, which of the following statements correctly describes the motion of the cart at a time $t > 5 \text{ s}$?
- (A) The cart is traveling in the $+x$ -direction and is slowing down.
 - (B) The cart is traveling in the $+x$ -direction and is speeding up.
 - (C) The cart is traveling in the $-x$ -direction and is slowing down.
 - (D) The cart is traveling in the $-x$ -direction and is speeding up.

U1

58. A solid metal ball and a hollow plastic ball of the same external radius are released from rest in a large vacuum chamber. When each has fallen 1 m, they both have the same
- (A) inertia
 - (B) speed
 - (C) momentum
 - (D) kinetic energy
 - (E) change in potential energy

59.

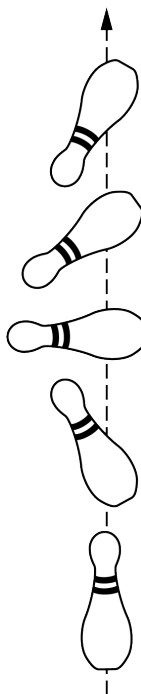
Speed	10 m/s	20 m/s	30 m/s
Braking Distance	6.1 m	23.9 m	53.5 m

To analyze the characteristics and performance of the brakes on a 1500 kg car, researchers collected the data shown in the table above. It shows the car's speed when the brakes are first applied and the corresponding braking distance required to stop the car. The magnitude of the average braking force on the car is most nearly

- (A) 75,000 N
- (B) 30,000 N
- (C) 12,000 N
- (D) 1600 N

U1

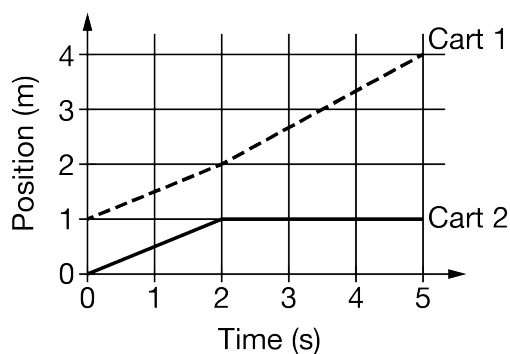
60.



A bowling pin is thrown vertically upward such that it rotates as it moves through the air, as shown in the figure. Initially, the center of mass of the bowling pin is moving upward with a speed v_i of $10 \frac{\text{m}}{\text{s}}$. The maximum height of the center of mass of the bowling pin is most nearly

- (A) $\frac{v_i^2}{2g}$
- (B) $\frac{2g}{v_i^2}$
- (C) The answer cannot be determined without knowing how long it takes the bowling pin to reach its maximum height.
- (D) The answer cannot be determined because an irregularly shaped object is too complex to treat as a single point mass.

61.

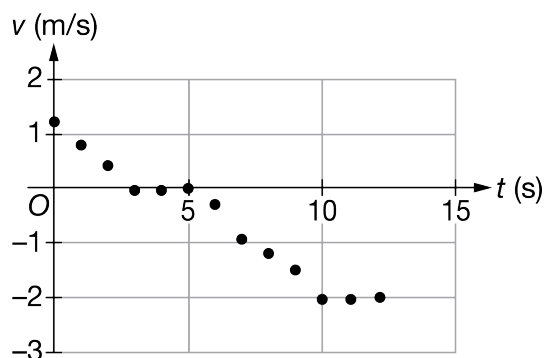


The graph above represents the position of two identical carts on a straight horizontal track as a function of time. Which of the following statements about the magnitude of the momentum of the two-cart system's center of mass, p_{com} , is true?

U1

- (A) p_{com} is constant from $t = 0$ s to $t = 5$ s.
(B) p_{com} is smaller after $t = 2$ s than it is before $t = 2$ s.
(C) p_{com} is larger after $t = 2$ s than it is before $t = 2$ s.
(D) p_{com} increases at a constant rate from $t = 0$ s to $t = 5$ s.

62.

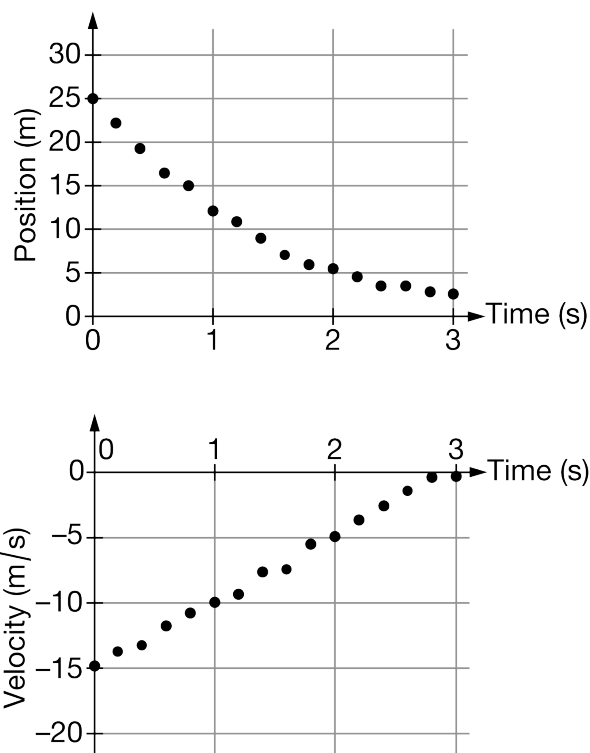


An engineer measures the velocity v of a remote-controlled cart on a straight track at regular time intervals. The data are shown in the graph above. During which of the following time intervals did the cart return to its position at time $t = 0$ s?

- (A) $3 \text{ s} \leq t < 5 \text{ s}$
(B) $5 \text{ s} \leq t < 7 \text{ s}$
(C) $7 \text{ s} \leq t < 10 \text{ s}$
(D) $10 \text{ s} \leq t < 12 \text{ s}$
63. A distant planet has an acceleration due to gravity of 4 m/s^2 near its surface. An object is released from rest from the top of a tall cliff on the planet, and the object lands at the bottom of the cliff in 20 seconds. A second object is then thrown upward from the edge of the same cliff with a speed of 4 m/s . The time it takes the second object to reach the bottom of the cliff is most nearly
- (A) 19 s
(B) 21 s
(C) 22 s
(D) 80 s
64. Two objects, X and Y , accelerate from rest with the same constant acceleration. Object X accelerates for twice the time as object Y . Which of the following is true of these objects at the end of their respective periods of acceleration?
- (A) Object X is moving at the same speed as object Y .
(B) Object X is moving four times faster than object Y .
(C) Object X has traveled the same distance as object Y .
(D) Object X has traveled twice as far as object Y .
(E) Object X has traveled four times as far as object Y .

U1

65.



The position and velocity of a car moving along a straight road are recorded as functions of time, as shown in the graphs above. Which of the following correctly describes the car's speed and acceleration?

U1

(A)

<u>Speed</u>	<u>Acceleration</u>
Increasing	Positive

(B)

<u>Speed</u>	<u>Acceleration</u>
Increasing	Negative

(C)

<u>Speed</u>	<u>Acceleration</u>
Decreasing	Positive

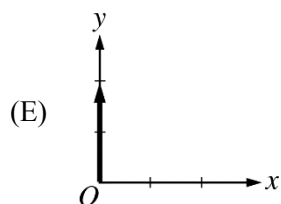
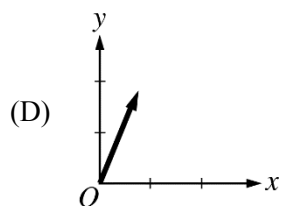
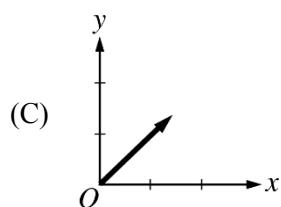
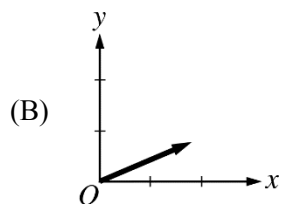
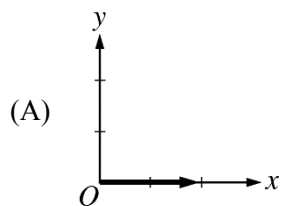
(D)

<u>Speed</u>	<u>Acceleration</u>
Decreasing	Negative

An object of mass m is initially at rest and free to move without friction in any direction in the xy -plane. A constant net force of magnitude F directed in the $+x$ direction acts on the object for 1 s. Immediately thereafter a constant net force of the same magnitude F directed in the $+y$ direction acts on the object for 1 s. After this, no forces act on the object.

66. Which of the following vectors could represent the velocity of the object at the end of 3 s, assuming the scales on the x and y axes are equal?

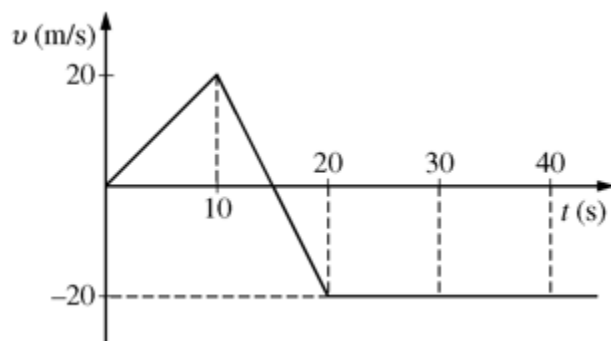
U1



67. Two students want to determine the speed at which a ball is released when thrown vertically upward into the air. One student throws the ball into the air while the other student measures the total time that the ball is in the air. The students use a meterstick to measure the release height of the ball. Which of the following equations should the students use to determine the speed at which the ball was released?
- (A) Use $y = y_0 + v_{y0}t + \frac{1}{2}a_y t^2$ from the moment in time in which the ball was released to the moment in time in which the ball reaches its highest point.
- (B) Use $y = y_0 + v_{y0}t + \frac{1}{2}a_y t^2$ from the moment in time in which the ball was released to the moment in time in which the ball hits the ground.
- (C) $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$ from the moment in time in which the ball was released to the moment in time in which the ball reaches its highest point.
- (D) $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$ from the moment in time in which the ball was released to the moment in time in which the ball hits the ground.

U1

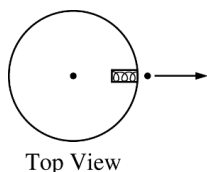
68.



An object begins at position $x = 0$ and moves one-dimensionally along the x -axis with a velocity v expressed as a function of time t according to the graph above. At what time does the object pass through $x = 0$ again?

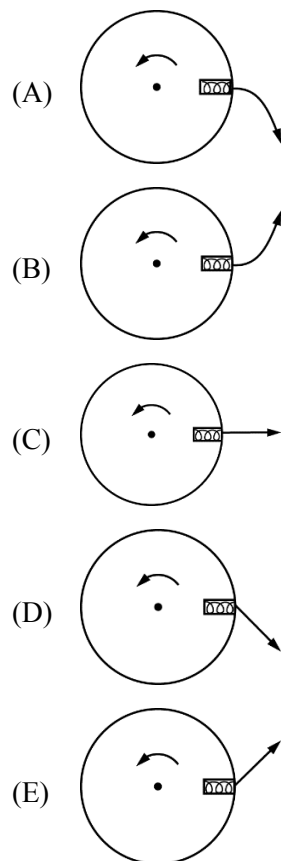
- (A) Between 10 s and 20 s
- (B) Between 20 s and 30 s
- (C) At 30 s exactly
- (D) Between 30 s and 40 s
- (E) After 40 s

69.



A compressed spring mounted on a disk can project a small ball. When the disk is not rotating, as shown in the top view above, the ball moves radially outward. The disk then rotates in a counterclockwise direction as seen from above, and the ball is projected outward at the instant the disk is in the position shown above. Which of the following best shows the subsequent path of the ball relative to the ground?

U1



70. An object is released from rest near the surface of a planet. The velocity of the object as a function of time is expressed in the following equation.

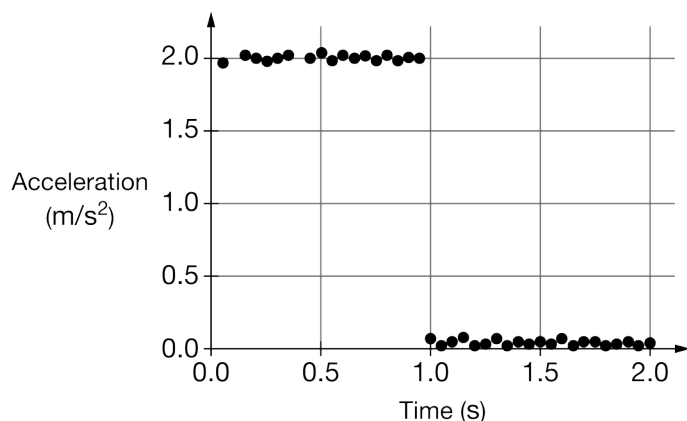
$$v_y = \left(-3 \frac{\text{m}}{\text{s}^2}\right)t$$

All frictional forces are considered to be negligible. What distance does the object fall 10 s after it is released from rest?

- (A) 3 m
- (B) 30 m
- (C) 150 m
- (D) 500 m

U1

71.



The graph shows the acceleration as a function of time for an object that is at rest at time $t = 0$ s. The distance traveled by the object between 0 and 2 s is most nearly

- (A) 1 m
- (B) 2 m
- (C) 3 m
- (D) 4 m

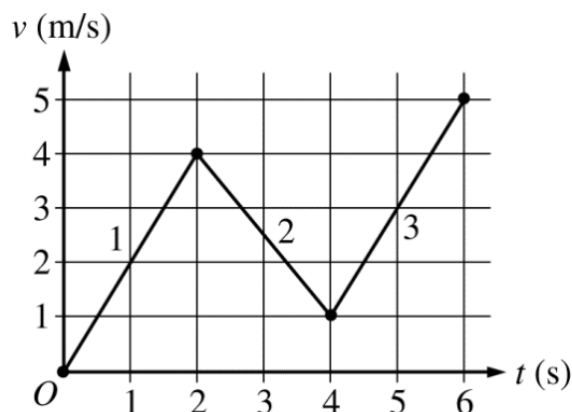
72. A car is moving with constant speed v_0 when a tree falls into the roadway at $t = 0$. It takes the driver a time T to react and apply the brakes. The car then slows with a constant acceleration of magnitude a . Which of the following equations correctly expresses the distance D traveled by the car from time $t = 0$ to the instant the car comes to rest?

- (A) $D = v_0 T + \frac{1}{2} a T^2$
- (B) $D = \frac{1}{2} a T^2 + \frac{v_0^2}{2a}$
- (C) $D = v_0 T + \frac{v_0^2}{2a}$
- (D) $D = v_0 T + \frac{1}{2} a T^2 + \frac{v_0^2}{2a}$

73. A car on a straight track is at position x_i and has initial speed v_i when it begins to slow down with constant acceleration of magnitude a . When the car reaches position x_f it has speed v_f . Which of the following is a correct expression for x_f ?

U1

- (A) $\frac{v_f^2 - v_i^2}{2a} + x_i$
- (B) $\frac{v_i^2 - v_f^2}{2a} + x_i$
- (C) $\frac{v_f^2 - v_i^2}{2a} - x_i$
- (D) $\frac{v_i^2 - v_f^2}{2a} - x_i$

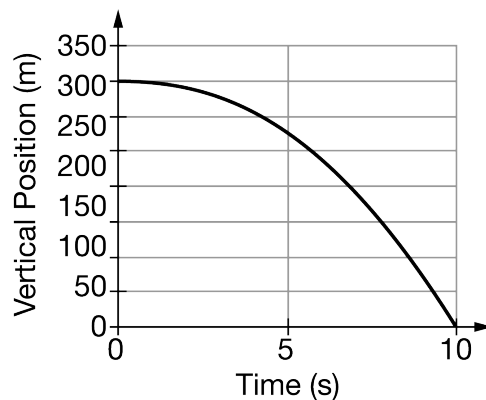


The graph above shows velocity v as a function of time t for a 0.50 kg object traveling along a straight line. The graph has three segments labeled 1, 2, and 3. A rope exerts a constant force of magnitude F_T on the object along its direction of motion the whole time. During segment 2 only, a frictional force of magnitude F_f is also exerted on the object.

74. Which of the following expressions correctly relates the magnitudes F_f and F_T ?
- (A) $F_f < F_T$
- (B) $F_f = F_T$
- (C) $F_T < F_f < 2F_T$
- (D) $F_f = 2F_T$
75. Which of the following correctly ranks the displacement Δx for the three segments of the object's motion?
- (A) $\Delta x_3 > \Delta x_2 > \Delta x_1 > 0$
- (B) $\Delta x_1 = \Delta x_2 = \Delta x_3 > 0$
- (C) $(\Delta x_1 = \Delta x_3) > \Delta x_2 > 0$
- (D) $(\Delta x_1 = \Delta x_3) > 0 > \Delta x_2$

U1

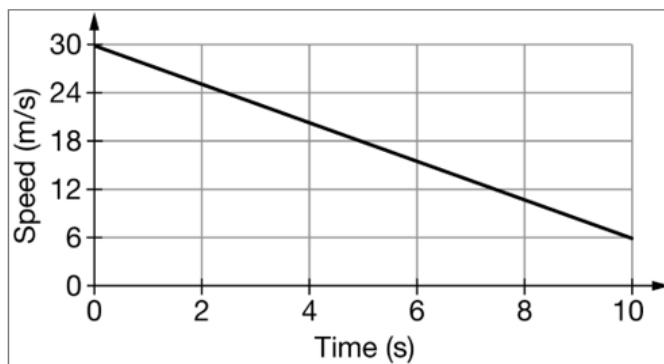
76.



An object is released from rest near the surface of a planet. The vertical position of the object as a function of time is shown in the graph. All frictional forces are considered to be negligible. The strength of the gravitational field is most nearly

- (A) $3 \frac{\text{N}}{\text{kg}}$
- (B) $6 \frac{\text{N}}{\text{kg}}$
- (C) $10 \frac{\text{N}}{\text{kg}}$
- (D) The strength of the gravitational field cannot be determined without applying Newton's law of universal gravitation.

77.

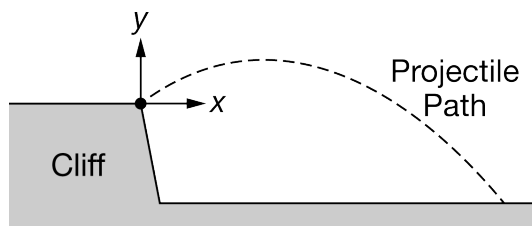


The graph above shows the speed of a truck as it moves along a straight, level road. How far does the truck travel in the 10 s time interval shown?

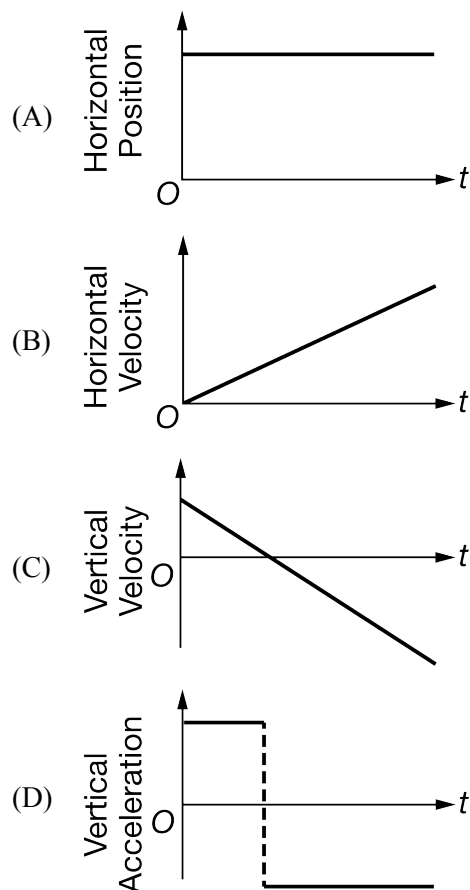
- (A) 120 m
- (B) 180 m
- (C) 240 m
- (D) 300 m

U1

78.



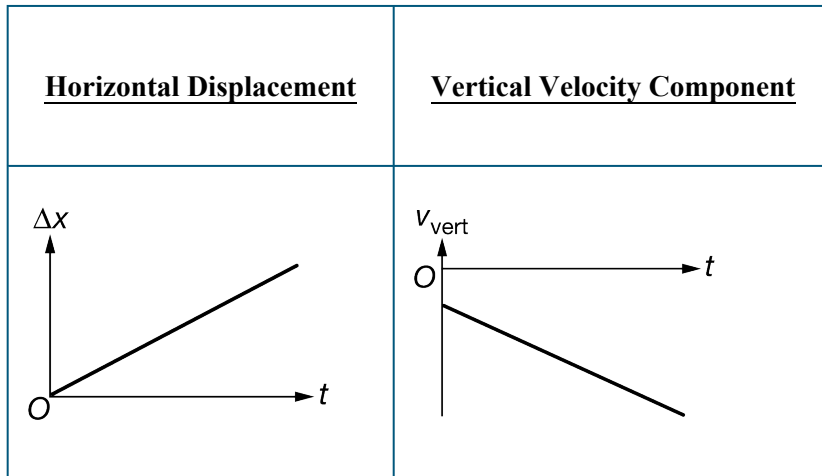
A projectile launched from a cliff at time $t = 0$ follows the path indicated by the dashed line in the above figure. The axes indicate the positive horizontal and vertical directions. Which of the following graphs could represent the projectile's motion?



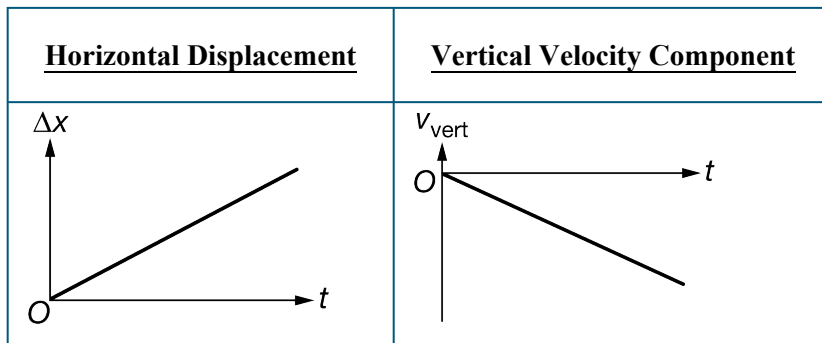
79. At time $t = 0$, a projectile is launched from the top of a cliff at an angle of 30° below the horizontal. Which of the following pairs of graphs best represents the horizontal displacement Δx and the vertical velocity component v_{vert} of the projectile as a function of time t ?

U1

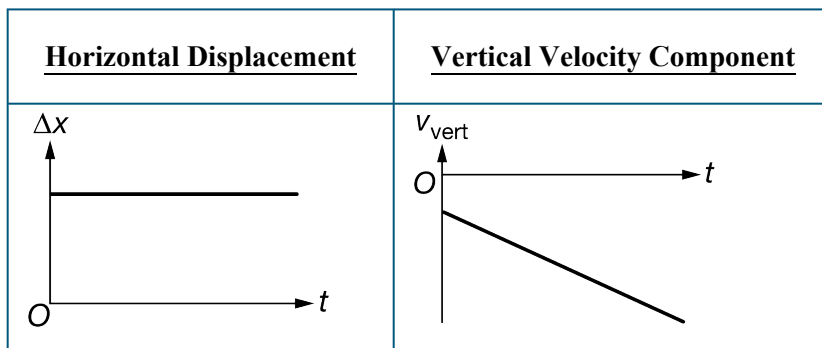
(A)



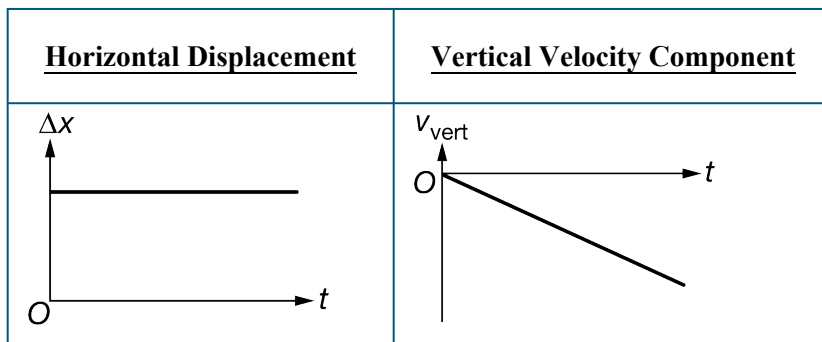
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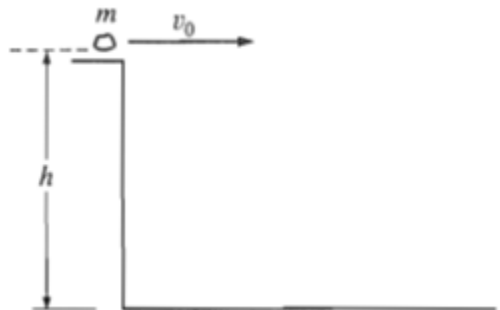
(C)



(D)



U1

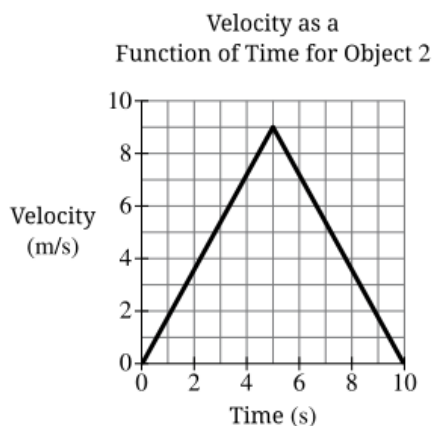
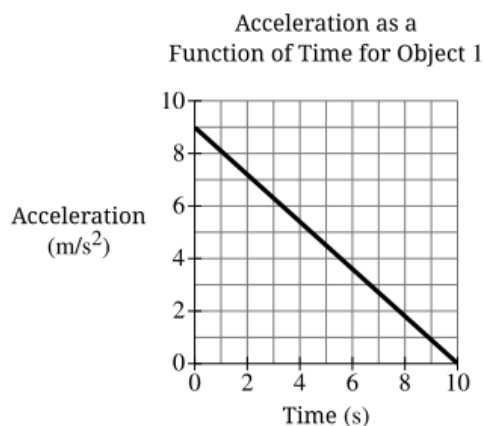


A rock of mass m is thrown horizontally off a building from a height h , as shown above. The speed of the rock as it leaves the thrower's hand at the edge of the building is v_0 .

80. How much time does it take the rock to travel from the edge of the building to the ground?

- (A) $\sqrt{h v_0}$
- (B) h/v_0
- (C) $h v_0/g$
- (D) $2h/g$
- (E) $\sqrt{2h/g}$

81.



Graphs of the motion of two identical objects 1 and 2 are shown in Figure 1 and Figure 2, respectively. The change in velocity of Object 1 from $t = 0$ to $t = 10$ s is Δv_1 . The change in velocity of Object 2 during the same time is Δv_2 . Which of the following correctly relates Δv_1 and Δv_2 ?

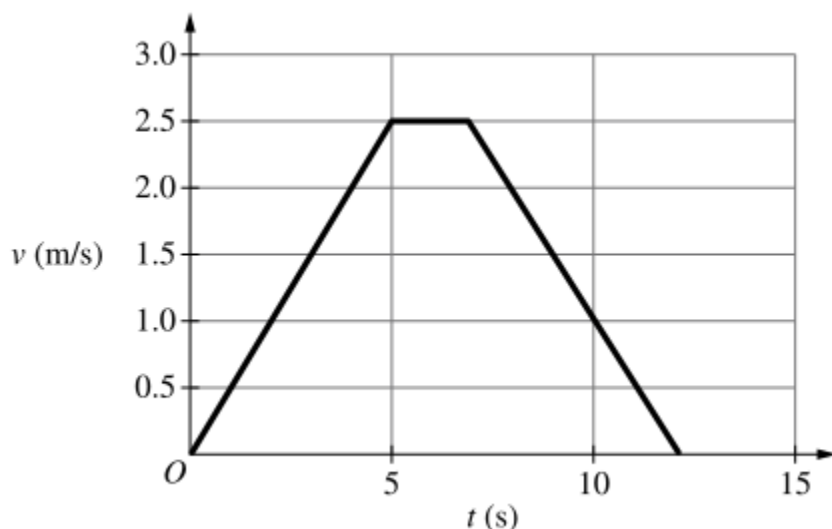
- (A) $\Delta v_1 > \Delta v_2 > 0$
- (B) $\Delta v_1 > (\Delta v_2 = 0)$
- (C) $(\Delta v_1 = \Delta v_2) > 0$
- (D) $\Delta v_1 = \Delta v_2 = 0$

U1

82. A model rocket is launched vertically from the ground. The rocket is launched from rest and accelerates upward with acceleration a for a time t . The engine then shuts off and the rocket continues to move upward. Which of the following is a correct expression that could be used to determine the maximum height above the ground reached by the rocket? Assume that the maximum height of the rocket is small compared to the radius of Earth.

- (A) $\frac{at^2}{2}$
(B) $\frac{a^2t^2}{2g}$
(C) $\frac{at^2}{2} + \frac{a^2t^2}{2g}$
(D) $\frac{at^2}{2} - \frac{a^2t^2}{2g}$

83.

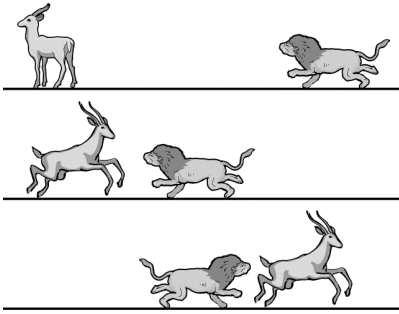


The motion of an object is shown in the velocity-time graph. Which of the following best describes the motion of the object?

- (A) The object is either speeding up or slowing down the entire time.
(B) The object starts and finishes at the same position.
(C) The object travels in the same direction for the entire time.
(D) The object undergoes positive acceleration the entire time.

U1

84.

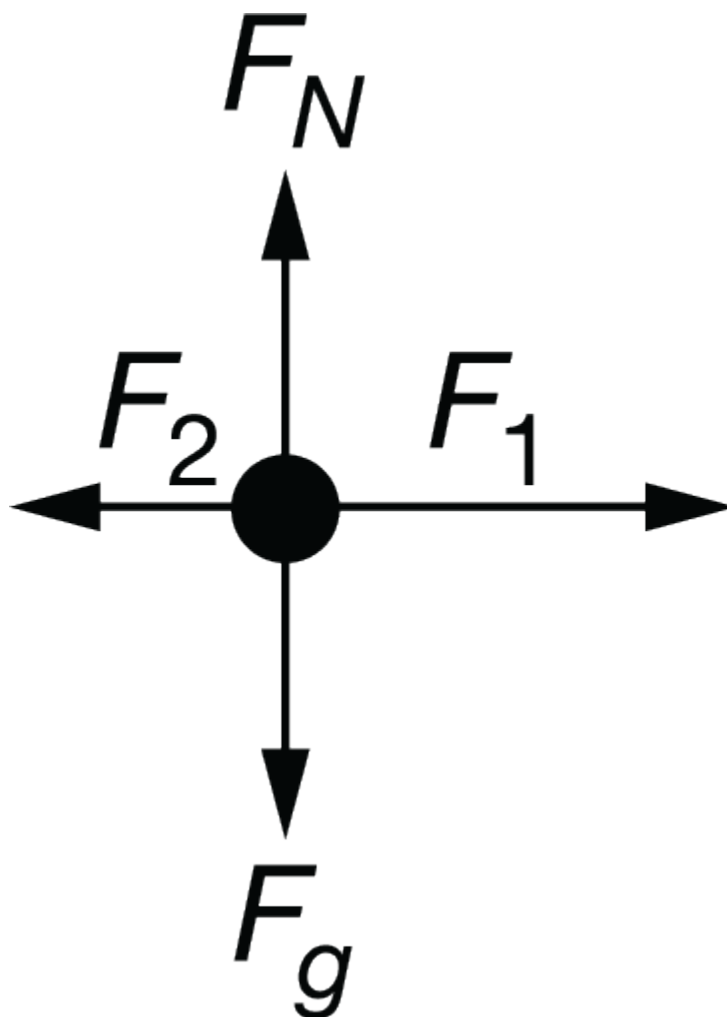


A lion is running at constant speed toward a gazelle that is standing still, as shown in the top figure above. After several seconds, the gazelle notices the lion and accelerates directly toward him, hoping to pass the lion and force him to reverse direction. As the gazelle accelerates toward and past the lion, the lion changes direction and accelerates in pursuit of the gazelle. The lion and the gazelle eventually each reach constant but different speeds. Which of the following sets of graphs shows a reasonable representation of the velocities of the lion and the gazelle as functions of time?

- (A)
- (B)
- (C)
- (D)

U1

85.

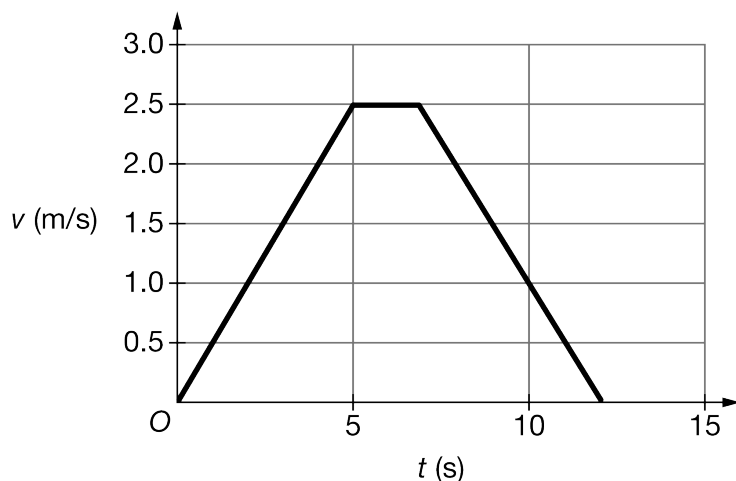


The diagram above shows the forces exerted on an object. Which of the following dot diagrams best represents the position of the object at equal time intervals?

- (A) 
- (B) 
- (C) 
- (D) 

U1

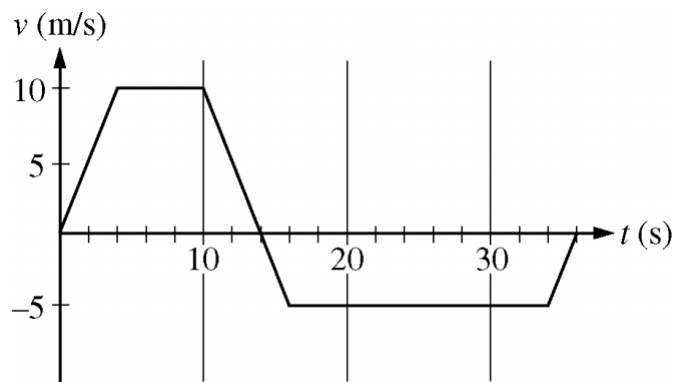
86.



The motion of an object is shown in the velocity-time graph. Which best describes the motion of the object?

- (A) The object is either speeding up or slowing down the entire time.
- (B) The object starts and finishes at the same position.
- (C) The object travels in the same direction for the entire time.
- (D) The object undergoes positive acceleration the entire time.

87.

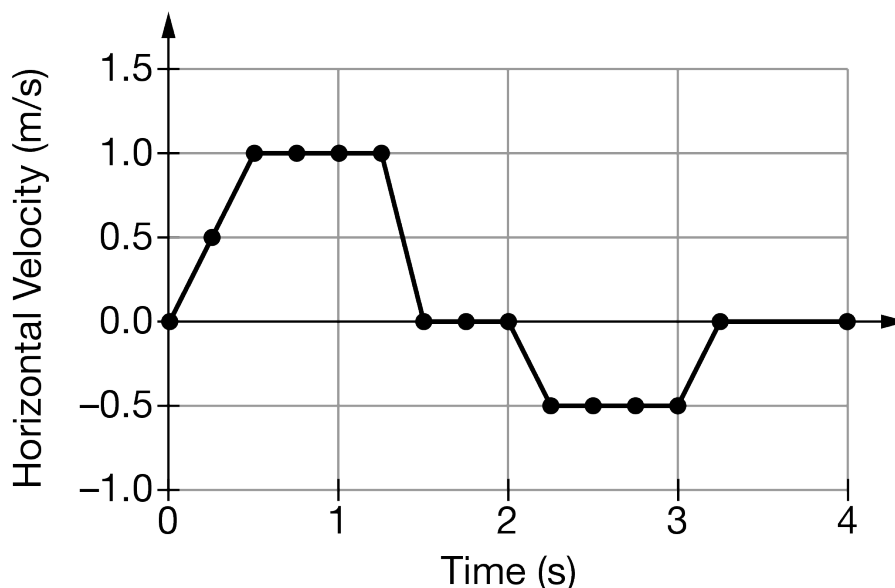


An object's velocity v as a function of time t is given in the graph above. Which of the following statements is true about the motion of the object?

- (A) The object is not moving from $t = 4$ s to $t = 10$ s.
- (B) The object's initial and final positions are the same.
- (C) The object is slowing down from $t = 14$ s to $t = 16$ s.
- (D) The average acceleration of the object from $t = 0$ s to $t = 4$ s is different from the acceleration from $t = 34$ s to $t = 36$ s.

U1

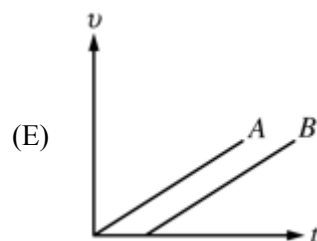
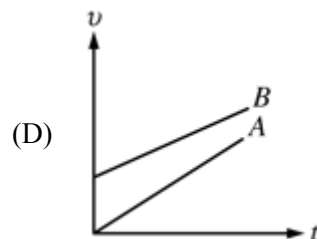
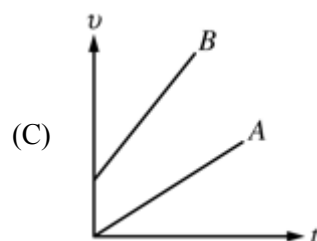
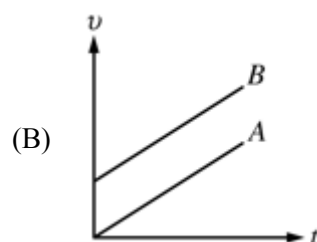
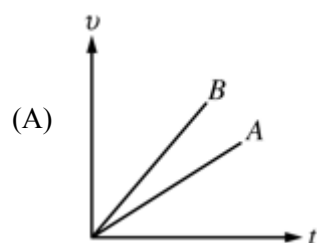
88.



A motion sensor is used to create the graph of a student's horizontal velocity as a function of time as the student moves toward and away from the sensor, as shown above. The positive direction is defined as the direction away from the sensor. Which of the following describes the student's final position x_f in relation to the starting position x_0 and the student's average horizontal acceleration a_x between 0.0 s and 3.0 s?

- (A) Position x_f is farther away from the sensor than x_0 , and a_x is positive.
 - (B) Position x_f is farther away from the sensor than x_0 , and a_x is negative.
 - (C) Position x_f is closer to the sensor than x_0 , and a_x is positive.
 - (D) Position x_f is closer to the sensor than x_0 , and a_x is negative.
89. At time $t = 0$ s, an object is moving to the right with a velocity v that can be modeled by the equation $v = (4.2 \text{ m/s}) - (1.4 \text{ m/s}^2)t$. At what time, if any, does the object change its direction of motion?
- (A) 0.33 s
 - (B) 3.0 s
 - (C) 5.6 s
 - (D) At no time; the object never changes direction.
90. Object A is released from rest at height h . At the same instant, object B is thrown downward from the same location. Which of the following graphs of speed of v as a function of time t is correct for the two objects?

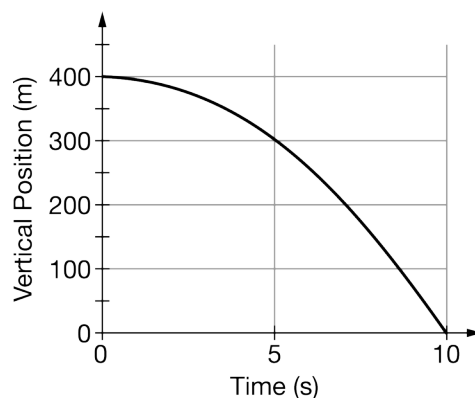
U1



91. An object is dropped near the surface of a planet such that the gravitational field at the object's location is $8 \frac{\text{N}}{\text{kg}}$. If the object is thrown upward at a speed of $20 \frac{\text{m}}{\text{s}}$, what is the position of the object in relation to the position in which the object was released and thrown upward after 3s ?
- (A) 96 m below the release position
 - (B) 36 m below the release position
 - (C) 15 m above the release position
 - (D) 24 m above the release position

U1

92.

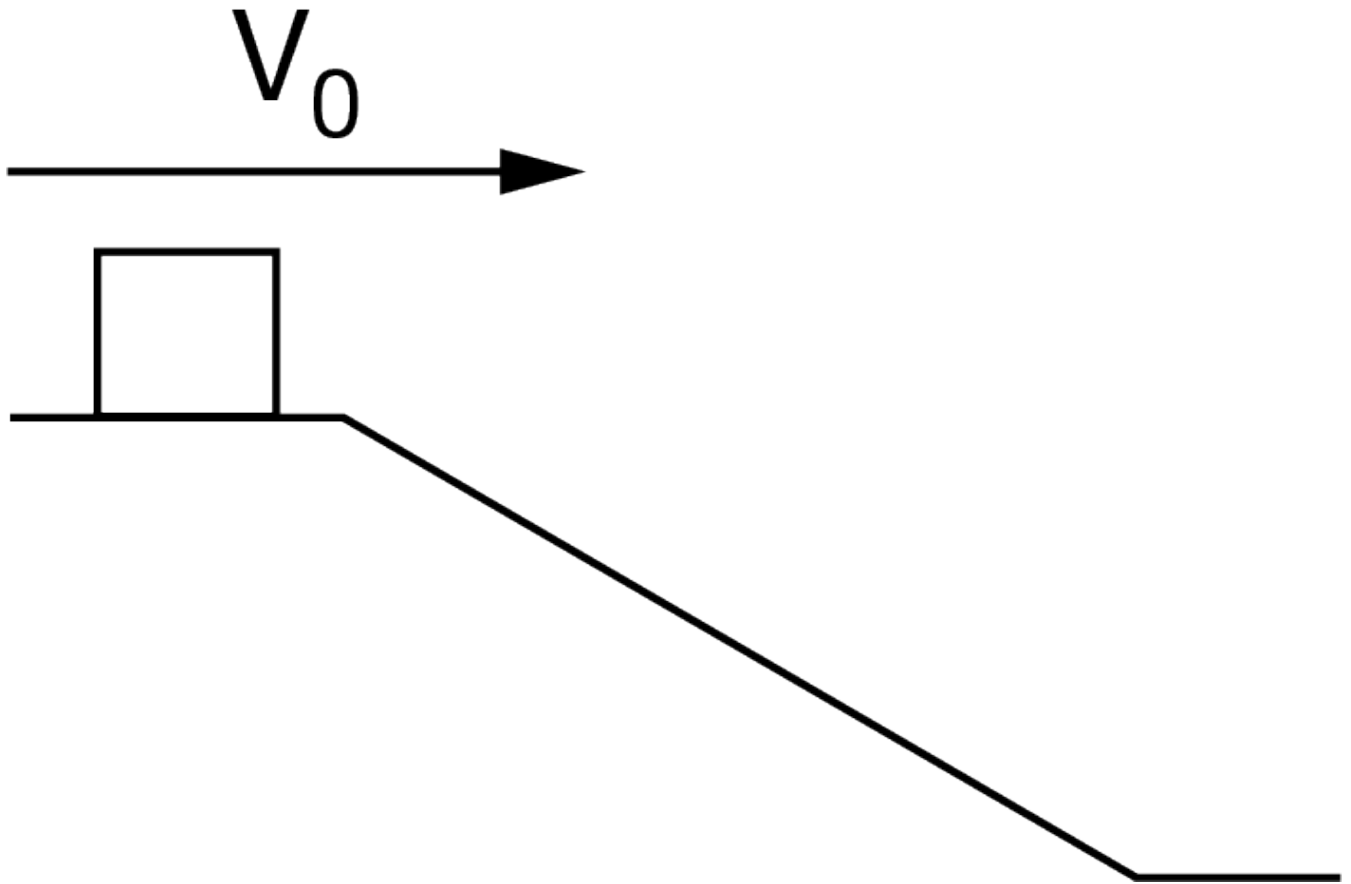


A 5 kg object is released from rest near the surface of a planet. The object's vertical position as a function of time is shown in the graph. Which of the following procedures can be used to determine the strength of the gravitational field on the planet?

- (A) Determine the average slope of the graph.
- (B) Use $v_y = v_{y0} + a_y t$. Use 10 s for t , $0 \frac{\text{m}}{\text{s}}$ for v_{y0} , and -40 m/s for v_{fy} . Solve for a_y .
- (C) Use $y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$. Use 10 s for t , $0 \frac{\text{m}}{\text{s}}$ for v_{y0} , 400 m for y_0 , and 0 m for y . Solve for a_y .
- (D) Use $\frac{F_g}{m} = g$. Use 5 kg for m and $10 \frac{\text{m}}{\text{s}^2}$ for g . Solve for F_g .

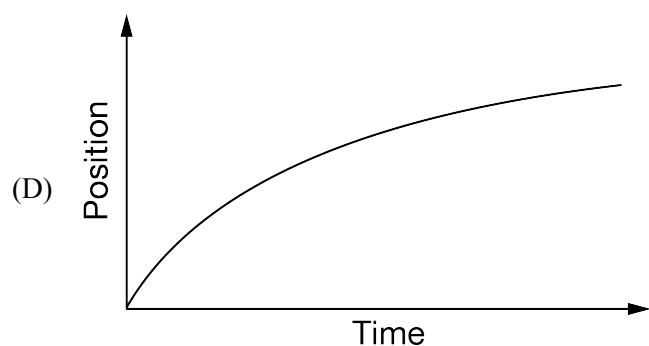
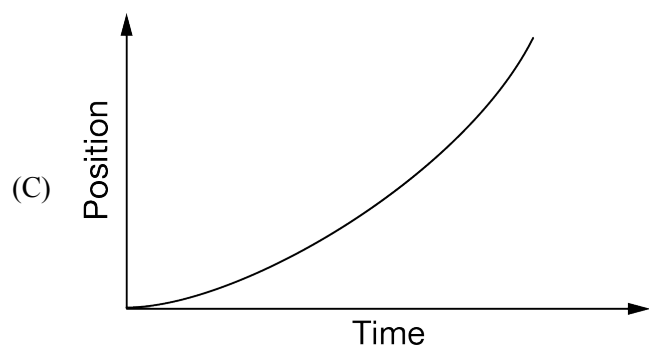
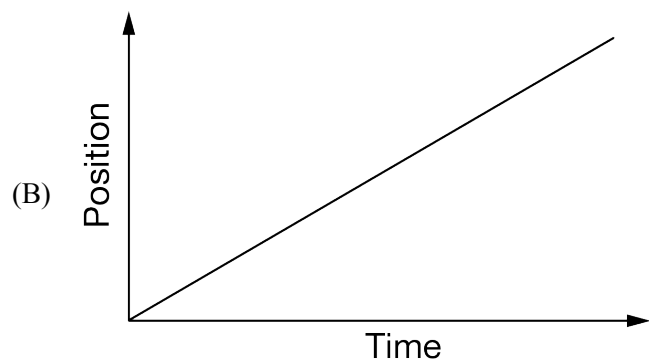
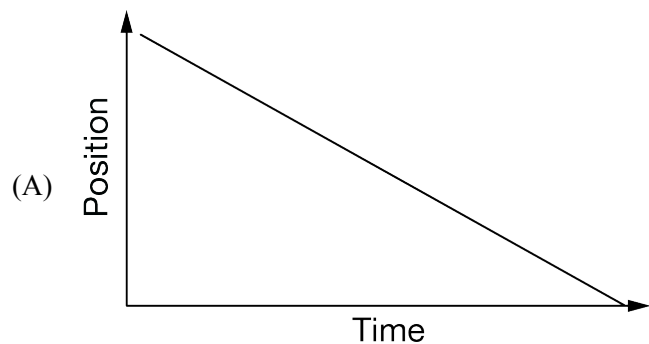
U1

93.



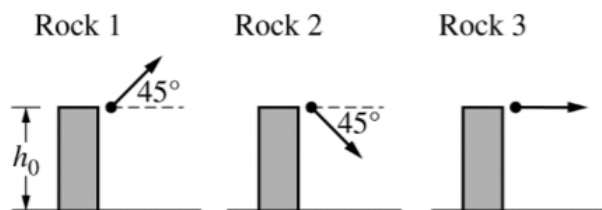
A block is moving horizontally with a speed of v_0 when it encounters a ramp, as shown above. Which of the following graphs best represents the position of the block measured from the top of the ramp as a function of time if friction is negligible?

U1



94. An object is dropped from a tall building at time $t = 0$ s and reaches the ground at $t = 10$ s. Which of the following correctly compares the distance Δx_{0-2} traveled by the object between $t = 0$ s and $t = 2$ s, the distance Δx_{4-6} traveled between $t = 4$ s and $t = 6$ s, and the distance Δx_{8-10} traveled between $t = 8$ s and $t = 10$ s?
- (A) $\Delta x_{0-2} > \Delta x_{4-6} > \Delta x_{8-10}$
- (B) $\Delta x_{0-2} = \Delta x_{4-6} = \Delta x_{8-10}$
- (C) $\Delta x_{8-10} > \Delta x_{4-6} > \Delta x_{0-2}$
- (D) $(\Delta x_{0-2} = \Delta x_{8-10}) > \Delta x_{4-6}$

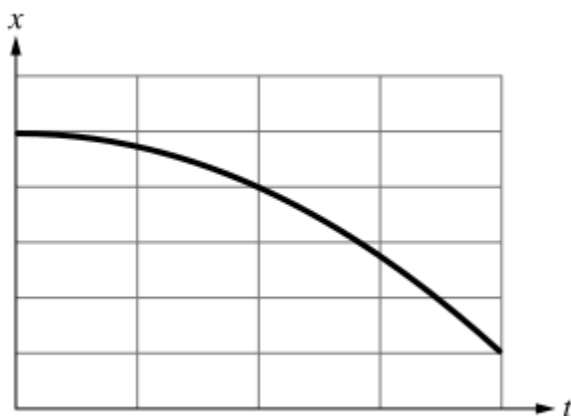
U1



Three identical rocks are launched with identical speeds from the top of a platform of height h_0 . The rocks are launched in the directions indicated above.

95. Which of the following correctly relates the magnitude v_y of the vertical component of the velocity of each rock immediately before it hits the ground?
- (A) $(v_{y1} = v_{y2}) > v_{y3}$
(B) $v_{y1} > v_{y3} > v_{y2}$
(C) $v_{y2} > v_{y3} > v_{y1}$
(D) $v_{y1} = v_{y2} = v_{y3}$

96.

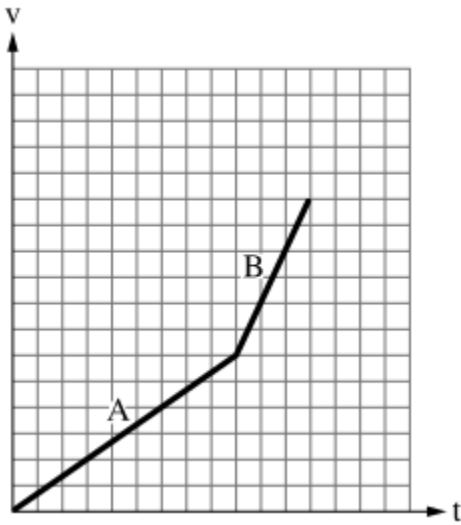


The graph shows the position x of an object as a function of time t as it moves along the x -axis. Which of the following claims about the motion of the object during the time interval shown is correct?

- (A) The object is moving with constant speed in the $-x$ -direction.
(B) The object is moving with constant speed in the $+x$ -direction.
(C) The velocity and acceleration are both in the $-x$ -direction.
(D) The velocity is in the $+x$ -direction and the acceleration is in the $-x$ -direction.

U1

97.

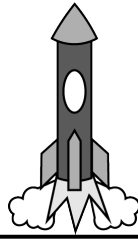


A cart starts from rest and moves to the right along a flat surface. The velocity of the cart as a function of time is represented in the graph. Which of the following quantities are the same for both segment A and segment B of the graph?

- (A) Acceleration
- (B) Displacement
- (C) Elapsed time
- (D) Average velocity

U1

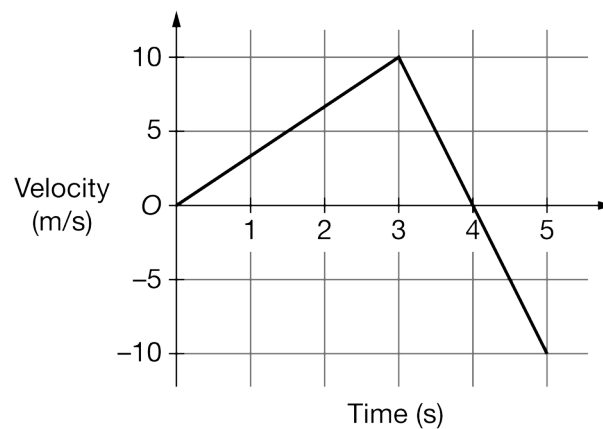
98. This question is a short free-response question. Show your work for each part of the question.



Time $t = 0$: Rocket Engine Ignites

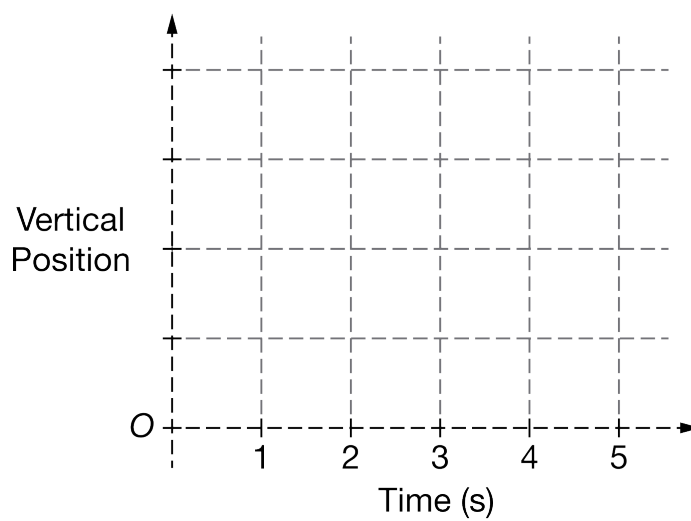
(7 points, suggested time 13 minutes)

A model rocket of constant mass 0.5 kg is launched vertically at time $t = 0$. The rocket's engine provides a constant force of magnitude F_0 for three seconds and then stops firing. The vertical velocity of the rocket as a function of time for the first 5 seconds of the flight is shown in the graph below.



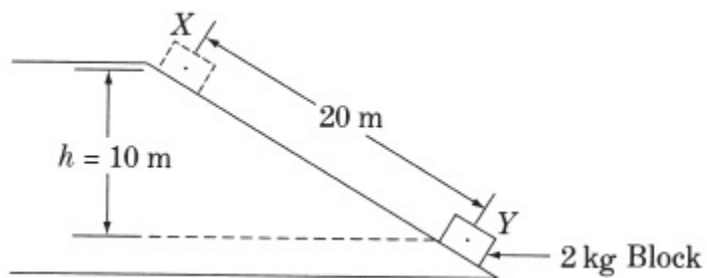
- (a) Calculate F_0 .
- (b) Calculate the time at which the rocket hits Earth after being launched.
- (c) On the axes below, sketch a graph of the vertical position of the rocket as a function of time for the first 5 seconds of the flight.

U1



99. A car initially at rest accelerates at $10 \frac{\text{m}}{\text{s}^2}$. The car's speed after it has traveled 25 meters is most nearly

- (A) $0.0 \frac{\text{m}}{\text{s}}$
 - (B) $10.0 \frac{\text{m}}{\text{s}}$
 - (C) $22.0 \frac{\text{m}}{\text{s}}$
 - (D) $25.0 \frac{\text{m}}{\text{s}}$
-



A 2 kg block, starting from rest, slides 20 m down a frictionless inclined plane from X to Y, dropping a vertical distance of 10 m as shown above.

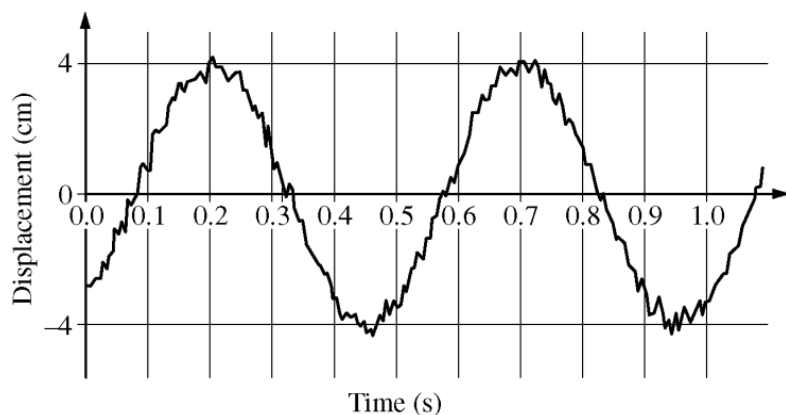
100. The speed of the block at point Y is most nearly

- (A) 7 m/s
 - (B) 10 m/s
 - (C) 14 m/s
 - (D) 20 m/s
 - (E) 100 m/s
-

U1

101. The magnitude of the net force on the block while it is sliding is most nearly

- (A) 0.1 N
- (B) 0.4 N
- (C) 2.5 N
- (D) 5.0 N
- (E) 10.0 N



A student sets an object attached to a spring into oscillatory motion and uses a position sensor to record the displacement of the object from equilibrium as a function of time. A portion of the recorded data is shown in the figure above.

102. The speed of the object at time $t = 0.65$ s is most nearly equal to which of the following?

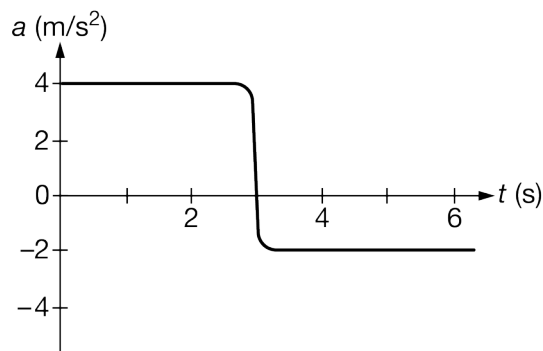
- (A) The value of the graph at 0.65 s
- (B) The slope of the line connecting the origin and the point on the graph at 0.65 s
- (C) The slope of the line connecting the point where the graph crosses the time axis near 0.57 s and the point on the graph at 0.65 s
- (D) The slope of the tangent to a best-fit sinusoidal curve at 0.65 s

103. The total distance traveled by the object between 0.35 s and 0.40 s is most nearly

- (A) 0 cm
- (B) 2 cm
- (C) 4 cm
- (D) 6 cm

U1

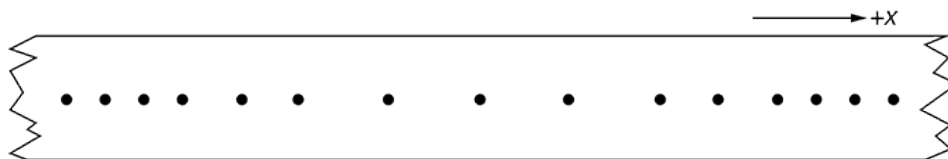
104.



The graph above shows the acceleration a of a car as a function of time t . Assuming the car starts from rest, the total displacement of the car from time $t = 0$ s to time $t = 6$ s is most nearly

- (A) 6 m
- (B) 9 m
- (C) 18 m
- (D) 45 m

105.

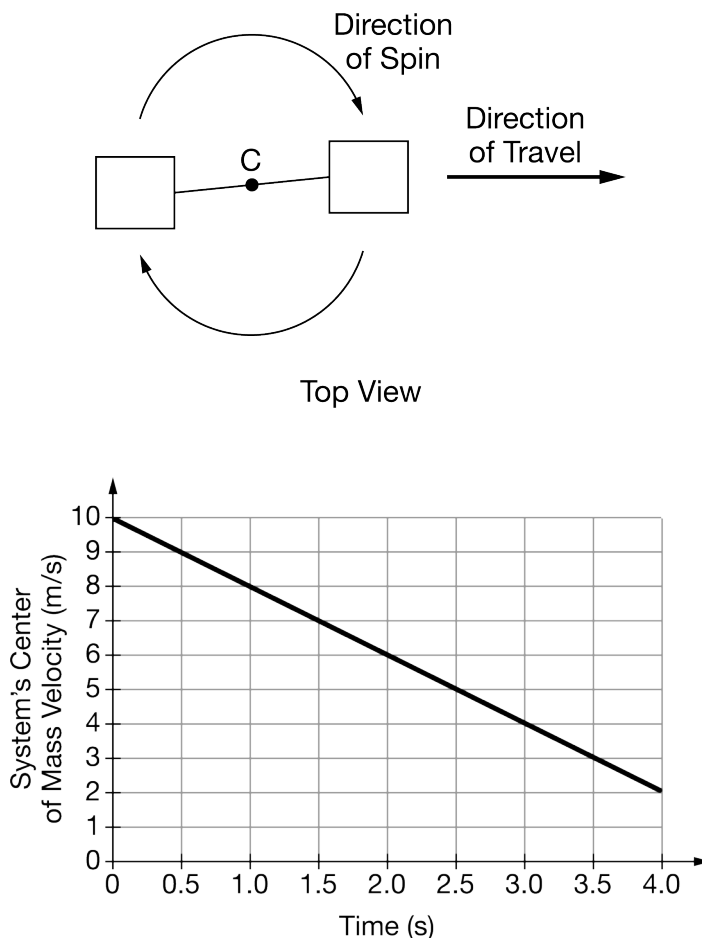


A student performs an experiment in which the horizontal position of a toy car is recorded on ticker tape from a device that places dots on the tape in equal time intervals. The series of dots in the figure represents the motion of an object moving from the negative direction to the positive direction along the horizontal direction. The time interval between each recorded dot is 1 s. Which of the following experiments could the student have conducted to create the data shown on the ticker tape?

- (A) A toy car that initially increases its speed, travels at a constant speed, and then increases its speed again.
- (B) A toy car that initially increases its speed, travels at a constant speed, and then decreases its speed.
- (C) A toy car that initially decreases its speed, stops, and then increases its speed.
- (D) A toy car that initially decreases its speed, travels at a constant speed, and then increases its speed.

U1

106.

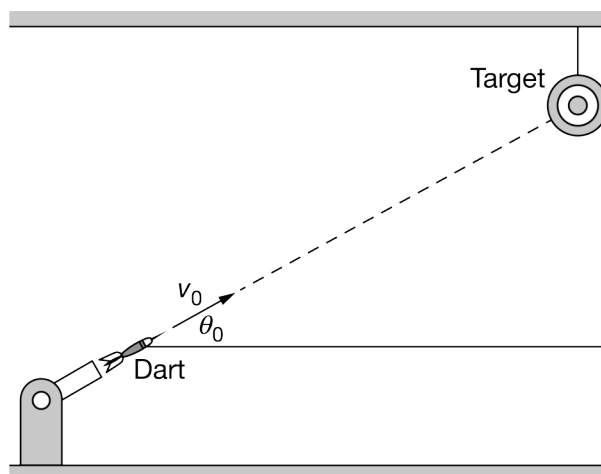


Two blocks are tied together with a string. They are thrown onto a layer of ice such that they spin around their center of mass C as they slide horizontally across the icy surface, as shown in the figure. A graph of the two-block system's velocity as a function of time is shown. Based on the graph, which of the following claims are correct about the system? Select two answers.

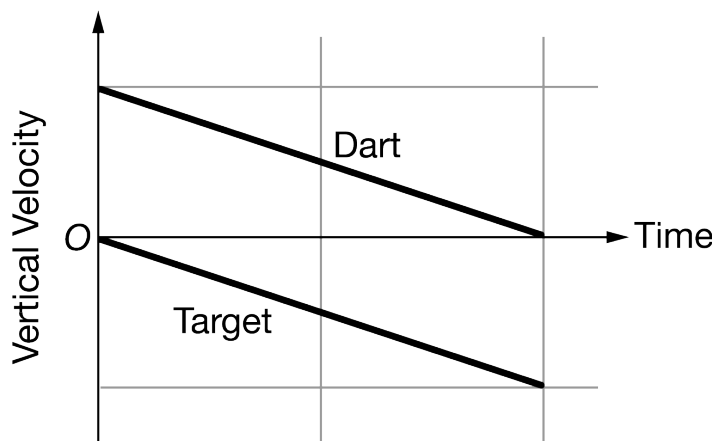
- ☐ A The total displacement of an individual block is zero because it travels in a perfect circle around the center of mass of the system.
- ☐ B The acceleration of the center of mass of the system is constant.
- ☐ C The change in velocity per unit of time for the system's center of mass changes as time increases.
- ☐ D The system's acceleration vector is opposite to the direction of the system's velocity vector.

U1

A student wants to launch a toy dart toward a target that hangs from a light string. At time $t = 0$, the dart is launched with an initial speed v_0 at an angle θ_0 above the horizontal ground. At the instant the dart is launched, the string is cut such that the target begins to fall straight down. The positive horizontal direction is considered to be to the right, and the positive vertical direction is considered to be up.



107.

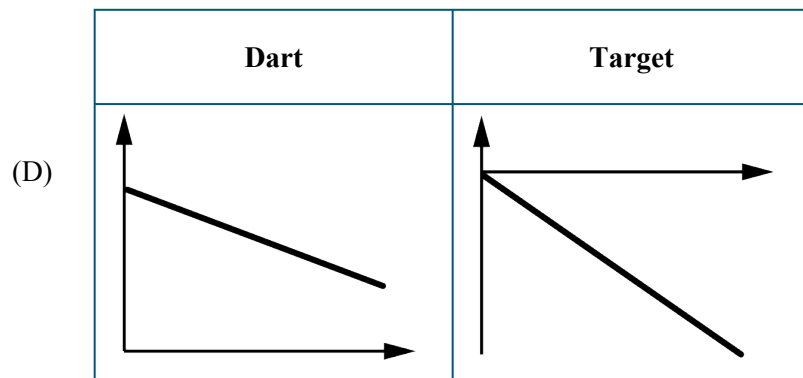
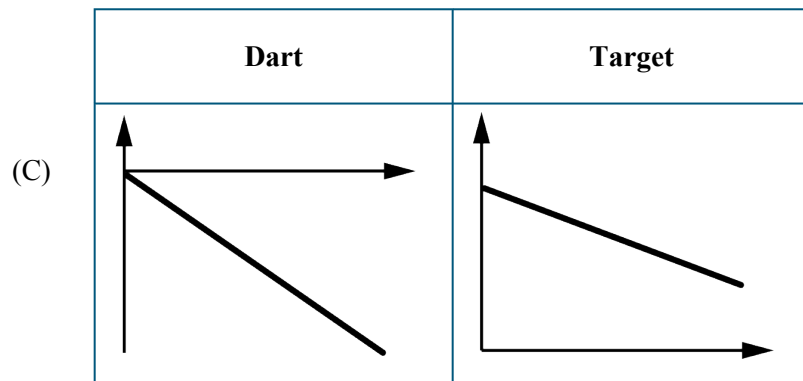
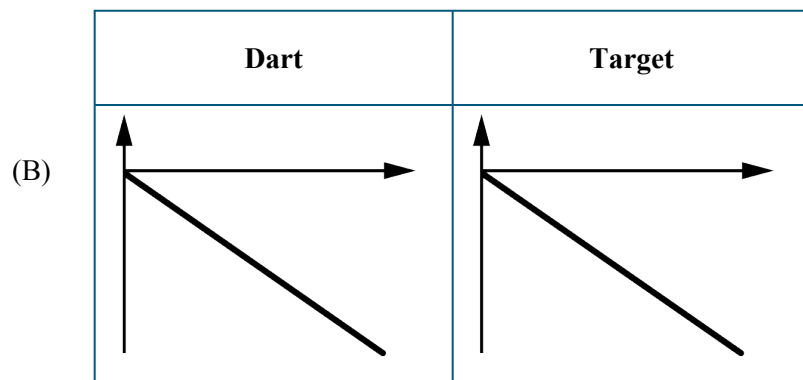
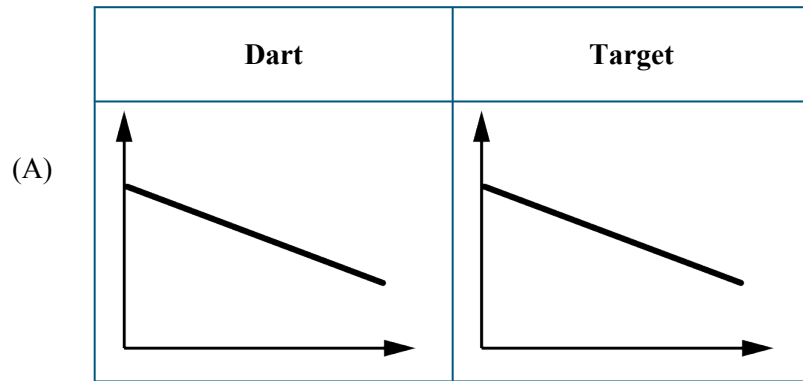


A student makes the necessary measurements to create the graph shown, which represents the vertical component of the velocity as a function of time for the dart and for the target from $t = 0$ until the instant the dart hits the target. At $t = 0$, the target is a vertical distance h above the dart. The curves for the dart and the target each have the same area between them and the horizontal axis. Both curves also have the same slope. Which of the following is the best method to determine the distance h from the graph?

- (A) Find the area under the curve for the dart's data.
- (B) Find twice the area under the curve for the dart's data.
- (C) Find the slope of the curve for the dart's data.
- (D) Find twice the slope of the curve for the dart's data.

108. Which of the following graphs could represent the vertical component of the velocity as a function of time for the dart and the target immediately after the dart is launched and the target begins to fall?

U1



U1

109.

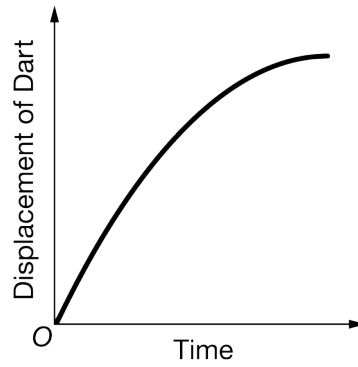


Figure 1

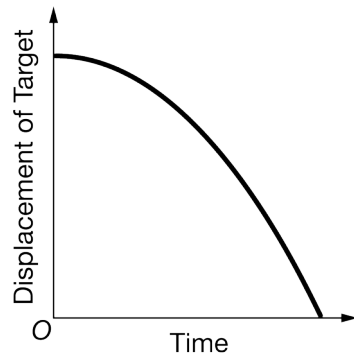


Figure 2

Figure 1 shows a displacement-versus-time graph for the dart. Figure 2 shows a displacement-versus-time graph for the target. For both graphs, the component of the displacement is not indicated.

Which displacement component, horizontal or vertical, is represented by the graph for each object?

U1

(A)

<u>Dart</u>	<u>Target</u>
Horizontal	Horizontal

(B)

<u>Dart</u>	<u>Target</u>
Vertical	Horizontal

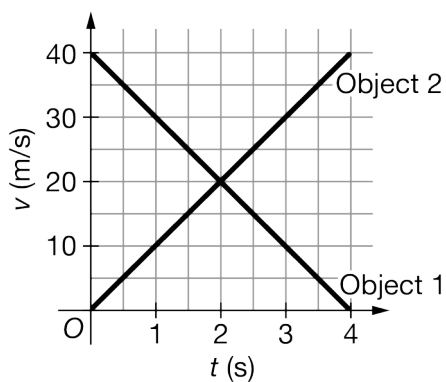
(C)

<u>Dart</u>	<u>Target</u>
Horizontal	Vertical

(D)

<u>Dart</u>	<u>Target</u>
Vertical	Vertical

110.



Object 1 and object 2 travel across a horizontal surface, and their horizontal velocity as a function of time is shown in the graph. Which of the following statements is correct about the two-object system?

- (A) The magnitude of acceleration for object 2 is greater than the magnitude of the acceleration of object 1.
- (B) After 2 s, object 1 travels a greater distance than object 2 travels.
- (C) Object 1 and object 2 are at the same horizontal position at 2 s.
- (D) After 4 s, the speed of object 1 is greater than the speed of object 2.

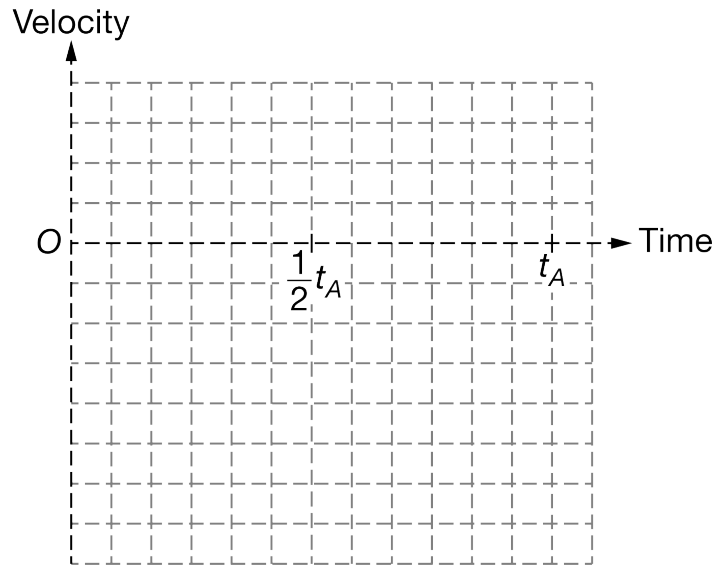
U1

111. This question is a short free-response question. Show your work for each part of the question.

(7 points, suggested time 13 minutes)

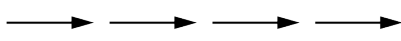
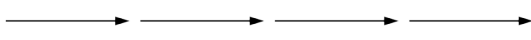
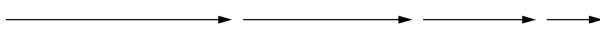
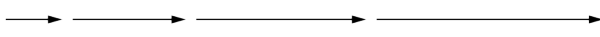
In an experiment two identical rocks are simultaneously thrown from the edge of a cliff a distance h_0 above the ground. Rock A is thrown vertically upward with speed v_0 and rock B is thrown vertically downward with speed v_0 . Rock A and rock B strike the ground at times t_A and t_B , respectively. Consider the positive vertical direction to be upward.

- (a) On the axes given below, sketch and label graphs of the velocity as a function of time for rock A and rock B. Label the time t_B . Times t_A and $\frac{1}{2}t_A$ are given on the graph.



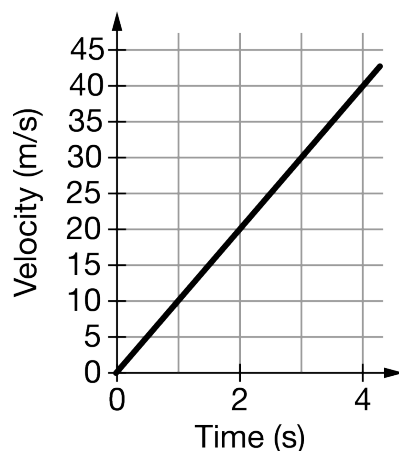
- (b) Rock B hits the ground at time t_B . Derive an equation for the time t_A it takes rock A to hit the ground in terms of v_0 , t_B , and physical constants, as appropriate.

112. A block slides across a flat, horizontal surface to the right. For each choice, the arrows represent velocity vectors of the block at successive intervals of time. Which of the following diagrams represents the situation in which the block loses kinetic energy?

- (A) 
- (B) 
- (C) 
- (D) 

U1

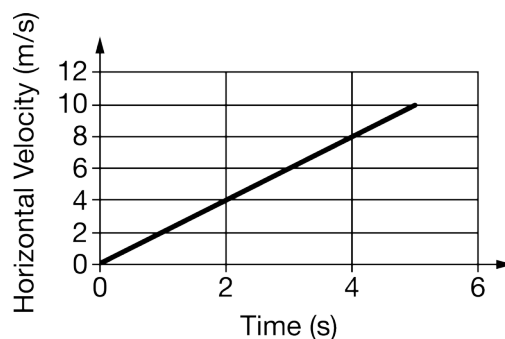
113.



A student drops a rock from the top of a cliff such that the rock falls downward toward Earth's surface in the absence of air resistance. The downward direction is considered to be the positive direction. The graph shows the rock's velocity as a function of time. Which of the following methods should be used to determine the total distance traveled by the rock after 4 s?

- (A) Use the maximum value of the curve, as it states that the rock traveled 40 meters in 1 second, which means that the rock traveled 160 meters in total.
- (B) Use the area under the curve, as it states that the rock traveled 80 meters in total.
- (C) Use the slope of the curve, as it states that the rock traveled $10 \frac{\text{m}}{\text{s}}$ in 1 second, which means that the rock traveled 40 meters in total.
- (D) The answer cannot be determined unless a graph of the object's position as a function of time is given.

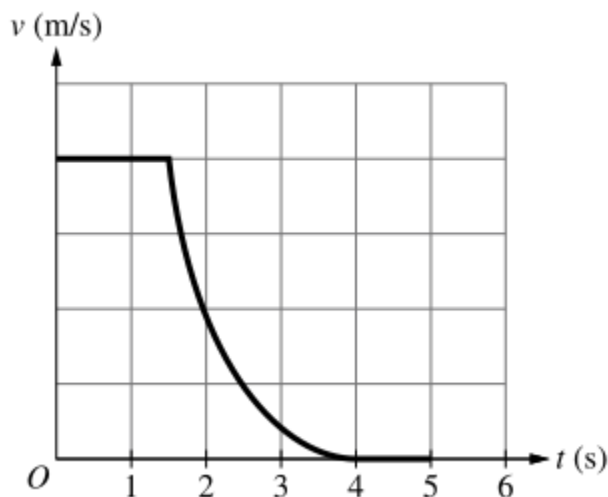
114.



An object travels along a straight line across a horizontal surface, and its motion is described by the velocity versus time graph shown in the figure. Which of the following methods will determine the total displacement of the object between 0 s and 5 s? Select two answers.

- ☐ A Finding the slope of the line between 0 s to 5 s
- ☐ B Dividing the change in velocity between 0 s to 5 s by the change in time
- ☐ C Finding the area bound by the horizontal axis and the curve from 0 s to 5 s
- ☐ D Using $\text{Average Speed} = \frac{\text{total distance}}{\text{total time}}$ and multiplying the average speed of $5 \frac{\text{m}}{\text{s}}$ by a total time of 5 s

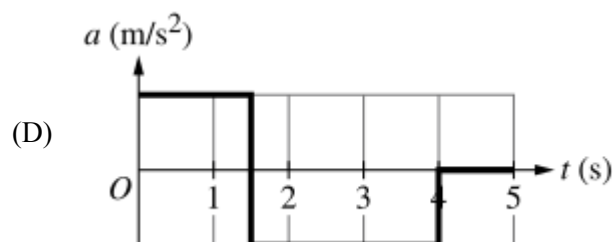
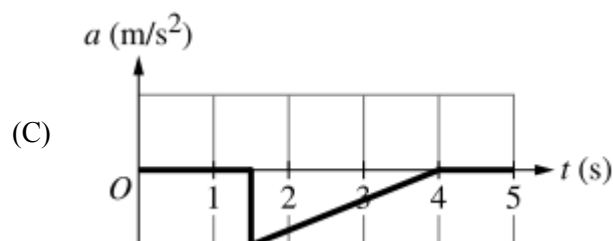
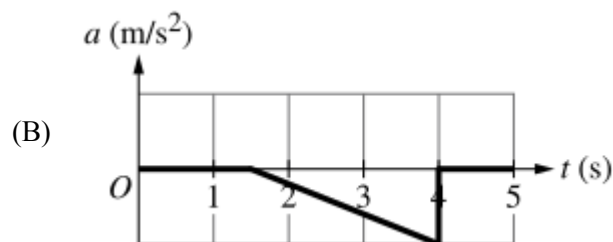
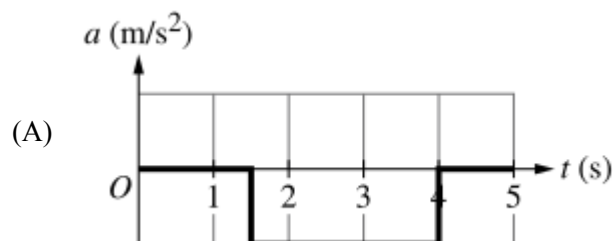
U1



At time $t = 0$, a car is moving in the positive direction along a straight road. The graph represents the car's velocity v as a function of time t . The car's position at $t = 0$ is $x = 0$.

115. The position of the car at times 1 s, 2 s, and 3 s are x_1 , x_2 , and x_3 , respectively. Which of the following correctly ranks x_1 , x_2 , and x_3 from greatest to least?
- (A) $x_1 > x_2 > x_3$
 - (B) $x_3 > x_2 > x_1$
 - (C) $x_2 > x_3 > x_1$
 - (D) $x_1 > x_3 > x_2$
116. Which of the following graphs could represent the acceleration a of the car as a function of time t ?

U1



117. A student wants to study the motion of an object that has a constant acceleration. Which of the following experiments could the student conduct to provide the best situations in which an object has a constant acceleration? Select two answers.

- ☐ A Release a ball from rest near Earth's surface.
- ☐ B Release a toy car from rest such that it travels along different looped sections of a track.
- ☐ C Launch a water-propelled rocket from rest such that it travels into the air and falls back to Earth's surface.
- ☐ D Release a cart from rest such that it travels down an incline of 40° with respect to the ground.

118. A student is riding on a bus that is traveling east at a constant speed. The student walks toward the front of the bus. A stationary observer outside the bus standing on the road claims that the student is moving faster than the bus. Is the claim correct? Why or why not?

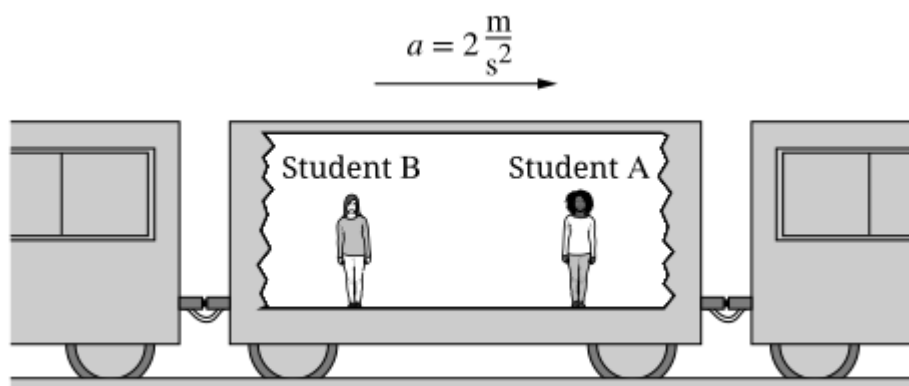
U1

- (A) No, because the student is included in the same frame of reference as the bus so the student and bus must be traveling at the same speed.
- (B) No, because the student, bus, and observer are all included in Earth's reference frame so the student and bus must be traveling at the same speed.
- (C) Yes, because according to the stationary observer, the student's velocity is the velocity of the bus plus the velocity of the student relative to the bus.
- (D) Yes, because according to the stationary observer, the student's velocity is the difference between the velocity of the bus and the velocity of the student relative to the bus.

119. Water in a river flows uniformly to the east with a speed of 5 m/s relative to the shore. A boat is traveling to the west at 10 m/s relative to the water. A passenger on the boat is walking at 5 m/s to the east, toward the back of the boat. Which of the following statements about the passenger's velocity is correct and gives the correct reason?

- (A) The passenger is moving at 20 m/s relative to the shore because the boat is moving at 5 m/s relative to the shore and the passenger is moving at 15 m/s relative to the water.
- (B) The passenger is moving at 20 m/s relative to the shore because the sum of all of the velocities (10 m/s , 5 m/s , and 5 m/s) is 20 m/s .
- (C) The passenger is stationary relative to the shore because the boat is moving at 5 m/s relative to the shore and the passenger is moving at 5 m/s relative to the boat in the opposite direction.
- (D) The passenger is stationary relative to the shore because the boat is moving at 15 m/s relative to the water and the passenger is moving at 15 m/s in the opposite direction relative to the water.

120.

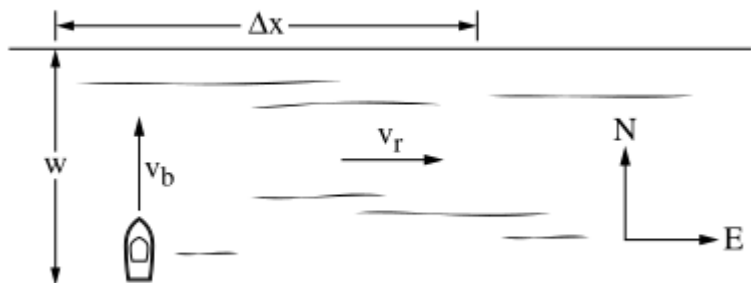


Students A and B are standing inside a train that is accelerating to the right with a magnitude of 2 m/s^2 . Student A throws a ball to the left toward Student B with a horizontal velocity of magnitude 4 m/s relative to the train. At the instant the ball leaves Student A's hand, the speed of the train relative to the ground is 10 m/s to the right. The ball takes 0.5 second to reach Student B. Immediately before the ball reaches Student B, the horizontal speed of the ball relative to the ground is

- (A) 7 m/s
- (B) 6 m/s
- (C) 5 m/s
- (D) 3 m/s

U1

121.



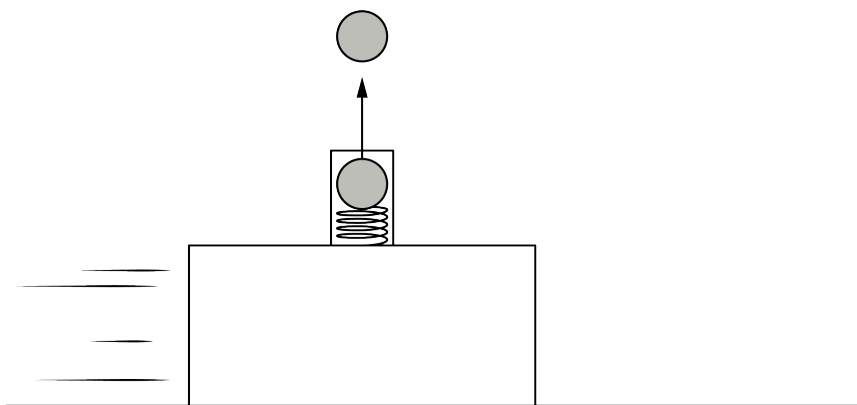
The water in a river moves uniformly to the east with speed v_r , as shown. A boat on the river starts from the south edge of the river and the boat's motor gives the boat a constant speed of v_b to the north relative to the water. The width of the river is w . When the boat reaches the north edge of the river, it has an eastward displacement of Δx relative to its starting position. Which of the following expressions is equal to v_r ?

- (A) $\sqrt{v_b^2 + 2a\Delta x}$
- (B) $\sqrt{w^2 + (\Delta x)^2}$
- (C) $\frac{\Delta x v_b}{w}$
- (D) $\tan^{-1}\left(\frac{w}{\Delta x}\right)$
122. A ball is thrown with an initial speed of 20 m/s at an angle of 60° to the ground. If air resistance is negligible, what is the ball's speed at the instant it reaches its maximum height from the ground?
- (A) Zero
- (B) 10 m/s
- (C) 14 m/s
- (D) 17 m/s
- (E) 20 m/s
123. A block is projected up a frictionless plane with an initial speed v_0 . The plane is inclined 30° above the horizontal. What is the approximate acceleration of the block at the instant that it reaches its highest point on the inclined plane?
- (A) Zero
- (B) 5 m/s^2 down the incline
- (C) 10 m/s^2 down the incline
- (D) 10 m/s^2 up the incline
- (E) It cannot be calculated without knowing the value of v_0

U1

124. A student throws a rock horizontally from the edge of a cliff that is 20 m high. The rock has an initial speed of 10 m/s. If air resistance is negligible, the distance from the base of the cliff to where the rock hits the level ground below the cliff is most nearly
- (A) 5 m
 - (B) 10 m
 - (C) 20 m
 - (D) 40 m
 - (E) 200 m

125.



A vertical spring launcher is attached to the top of a block and a ball is placed in the launcher, as shown in the figure. While the block slides at constant speed to the right across a horizontal surface with negligible friction between the block and the surface, the ball is launched upward. When the ball reaches its maximum height, what will be the position of the ball relative to the spring launcher?

- (A) Above and to the right of the spring launcher
- (B) Directly above the spring launcher
- (C) Above and to the left of the spring launcher
- (D) The relative position of the ball depends on the horizontal speed of the block.

U1

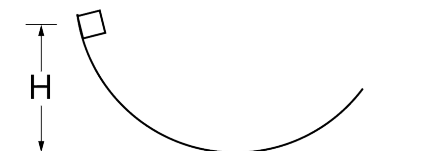


Figure 1



Figure 2

Note: Figures not drawn to scale.

The figures above show a small block of mass 0.20 kg on a track in the shape of a circular arc. The block is released from rest at a height H above the floor, as shown in Figure 1. The block slides along the track with negligible friction and leaves it at a height of 0.40 m above the floor and a speed of 3.0 m/s at a 30° angle, as shown in Figure 2.

126. After the block leaves the track, what is the block's speed when it reaches the highest point of its motion?

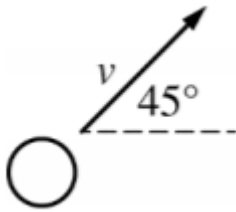
- (A) 0
- (B) 1.5 m/s
- (C) 2.6 m/s
- (D) 3.0 m/s

127. Two people are in a boat that is capable of a maximum speed of 5 km/hr in still water. The people wish to cross a river 1 km wide to a point directly across from their starting point. If the speed of the water in the river is 4 km/hr , what is the minimum time required for the crossing?

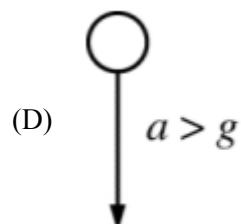
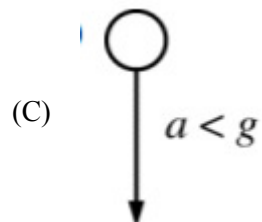
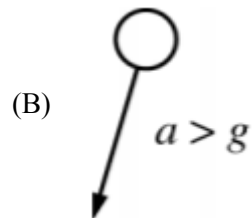
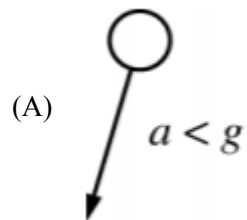
- (A) $\frac{1}{9} \text{ hr}$
- (B) $\frac{1}{5} \text{ hr}$
- (C) $\frac{1}{4} \text{ hr}$
- (D) $\frac{1}{3} \text{ hr}$

U1

128.

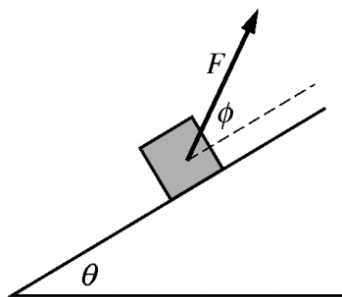


A hollow plastic ball is projected into the air. There is significant air resistance opposing the ball's motion, so the magnitude of the ball's acceleration is not equal to g . At time t , the ball is moving up and to the right at an angle of 45° to the horizontal, as shown above. Which of the following best shows the magnitude a and the direction of the ball's acceleration at time t ?



U1

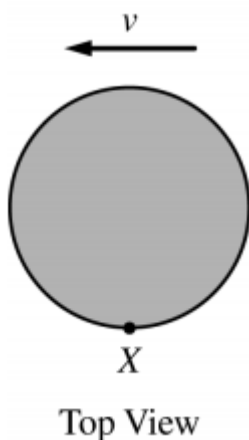
129.



A box of mass m is on a rough inclined plane that is at an angle θ with the horizontal. A force of magnitude F at an angle Φ with the plane is exerted on the block, as shown above. As the block moves up the plane, there is a frictional force between the box and the plane of magnitude f . What is the magnitude of the net force acting on the box?

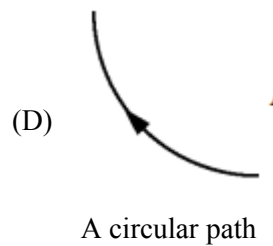
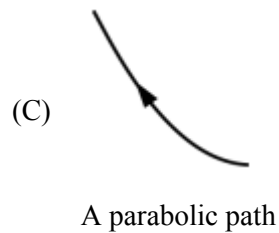
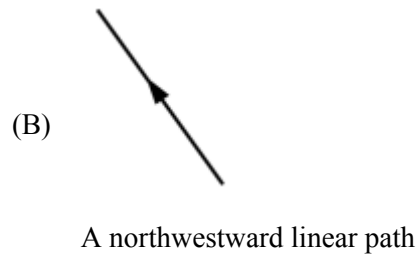
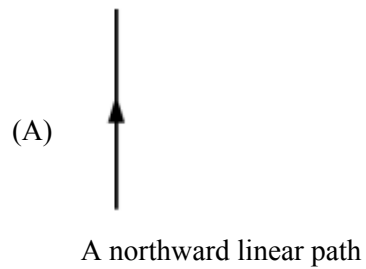
- (A) $F \sin \Phi - mg \cos \theta - f$
- (B) $F \cos(\Phi + \theta) + mg \sin \theta - f$
- (C) $F \cos \Phi - mg \sin \theta - f$
- (D) $F \cos(\Phi + \theta) - mg \sin \theta - f$

130.



A stone disk is sliding on frictionless ice to the west with speed v , as shown in the figure above. As the disk slides by, a child uses a rubber mallet to hit the disk at point X, exerting a force directly toward the center of the disk. The child hits point X every half second for about 10 s, changing the trajectory of the disk but not causing it to rotate. Which of the following most closely approximates the path of the disk while the child is hitting it?

U1

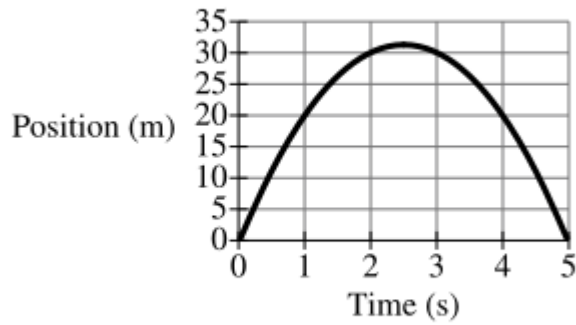


131. A projectile is launched from level ground with initial velocity v_0 , at an angle θ above horizontal. Which of the following is an expression for the total time the projectile is in the air?

- (A) $t = \frac{v_0 \cos \theta}{g}$
- (B) $t = \frac{2v_0 \cos \theta}{g}$
- (C) $t = \frac{v_0 \sin \theta}{g}$
- (D) $t = \frac{2v_0 \sin \theta}{g}$

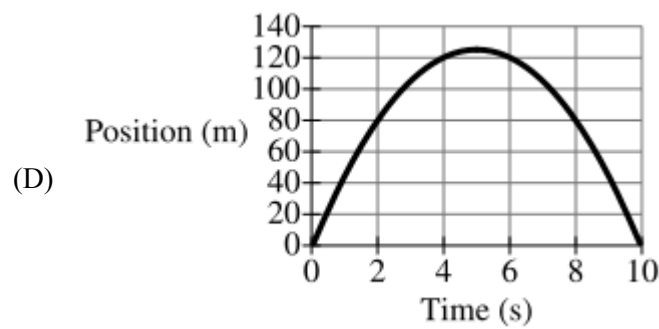
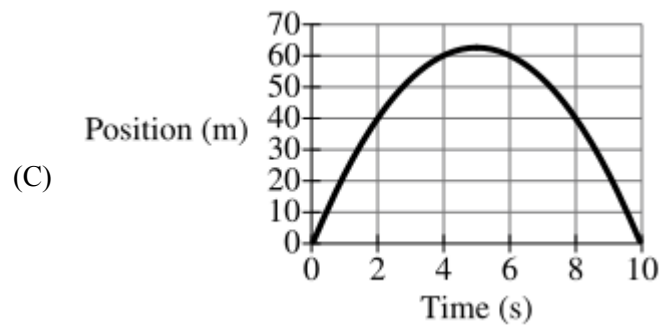
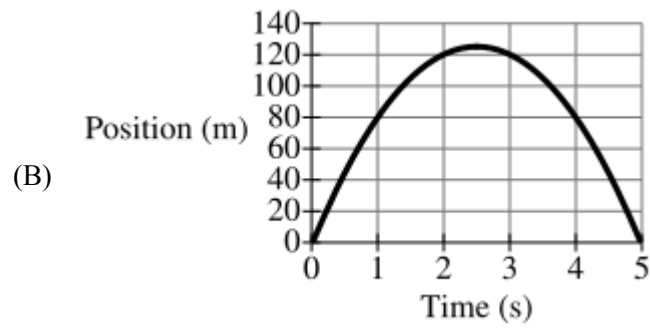
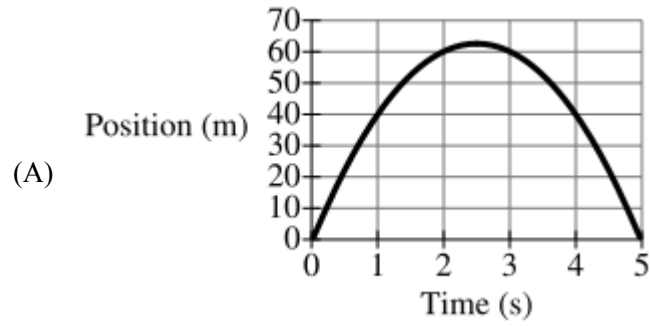
U1

132.



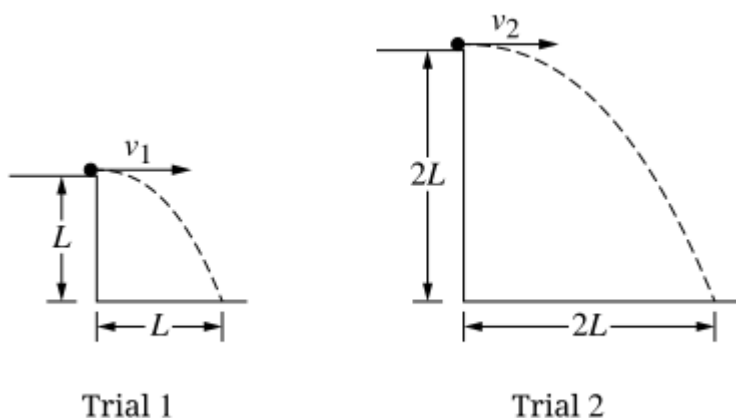
The graph shows the vertical position as a function of time for an object launched from the ground. Which of the following graphs correctly shows the vertical position of the object as a function of time if the object was launched at the same angle but with twice the initial speed?

U1



U1

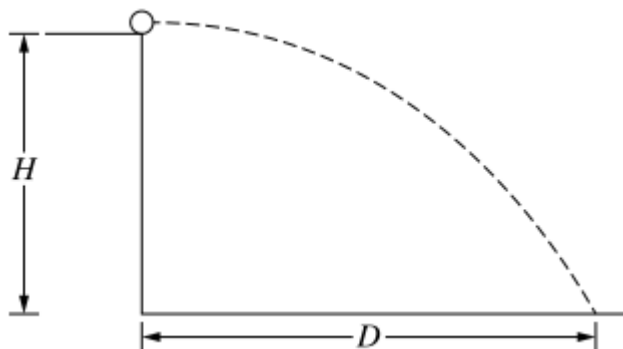
133.



In Trial 1, a sphere is projected horizontally from an initial height L and travels horizontally a distance L before reaching the floor. The sphere's initial velocity for this trial is v_1 and the time required for the sphere to reach the floor is t_1 . In Trial 2, the sphere is projected horizontally from an initial height $2L$ and travels horizontally a distance $2L$ before reaching the floor. The sphere's initial velocity for this trial is v_2 and the time required for the sphere to reach the floor is t_2 . Which of the following pairs of equations correctly relates t_2 and v_2 to t_1 and v_1 ?

- (A) $t_2 = \sqrt{2}t_1$ and $v_2 = \sqrt{2}v_1$
- (B) $t_2 = \sqrt{2}t_1$ and $v_2 = 2v_1$
- (C) $t_2 = 2t_1$ and $v_2 = \sqrt{2}v_1$
- (D) $t_2 = 2t_1$ and $v_2 = 2v_1$

134.

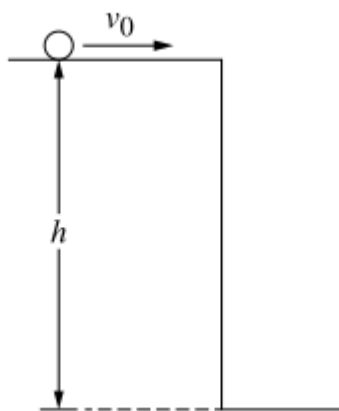


A ball is projected horizontally from a table with height H above the ground. The ball lands after a time t_1 , hitting the ground a horizontal distance D away from the edge of the table, as shown. Which of the following is a correct expression for the instantaneous speed of the ball immediately before it reaches the ground?

U1

- (A) $\sqrt{\left(\frac{D}{t_1}\right)^2 + (gt_1)^2}$
- (B) $\frac{D^2}{2gt_1^2}$
- (C) $\frac{H}{t_1}$
- (D) $\frac{D}{t_1}$

135.

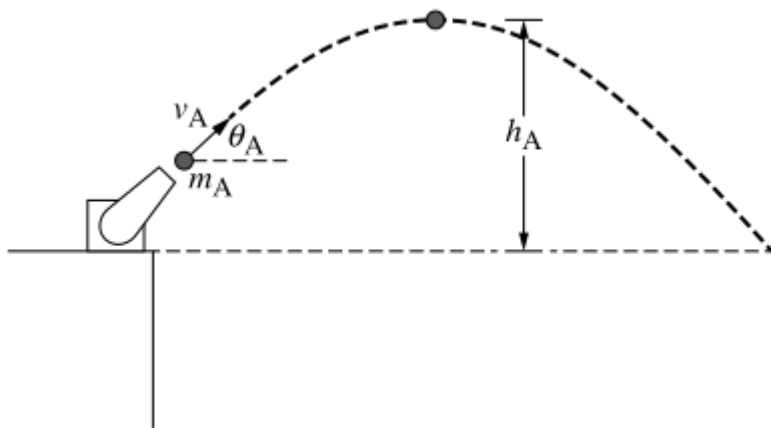


A projectile is launched horizontally off a cliff of known height h with speed v_0 and reaches the ground a time t after being launched. Another identical projectile is launched horizontally off the same cliff with speed $2v_0$. How long does it take the second projectile to reach the ground?

- (A) $\frac{t}{2}$
- (B) t
- (C) $2t$
- (D) $4t$

U1

136.

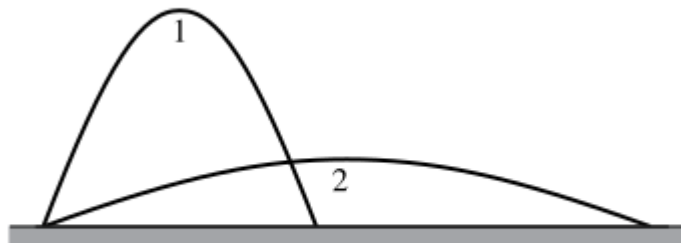


A student launches a marble of mass m_A with initial speed v_A at an angle θ_A above the horizontal, as shown in the diagram. The marble reaches a maximum height of h_A above the tabletop. The student launches another marble with a mass $\frac{1}{2}m_A$ at the same speed and angle as the first marble. Will the maximum height of the second marble be equal to h_A ? Why or why not?

- (A) Yes, because the initial vertical components of the velocity are the same for both marbles.
 - (B) Yes, because the second marble has half the mass of the first marble, but it will experience twice the vertical acceleration.
 - (C) No, because the less massive marble will have a smaller acceleration.
 - (D) No, because the less massive marble will have a smaller initial vertical velocity.
137. A ball is projected horizontally with initial speed v_0 from a platform of height h on Earth. The ball's speed immediately before touching the ground is v_e . The experiment is repeated on the Moon so that the ball is projected horizontally with the same speed from the same height. The acceleration due to gravity is less on the Moon than on the Earth. The ball's speed just before touching the ground is v_m . Which speed is greater, and why?
- (A) v_m , because the ball on the Moon will travel a greater horizontal distance before reaching the ground
 - (B) v_m , because the ball on the Moon has less acceleration to slow it down when it is on the Moon
 - (C) v_e , because the ball on the Moon achieves a smaller vertical component of velocity before reaching the ground
 - (D) v_e , because the ball on the moon travels less distance horizontally each second than the ball on the Earth

U1

138.

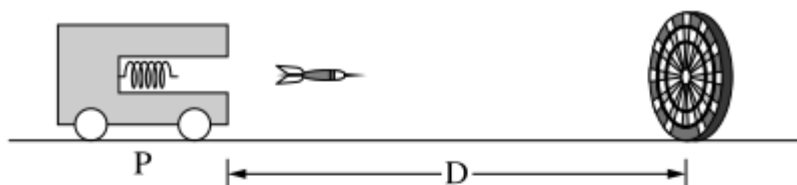


Projectiles 1 and 2 are launched from level ground at the same time and follow the trajectories shown in the figure. Which one of the projectiles, if either, returns to the ground first, and why?

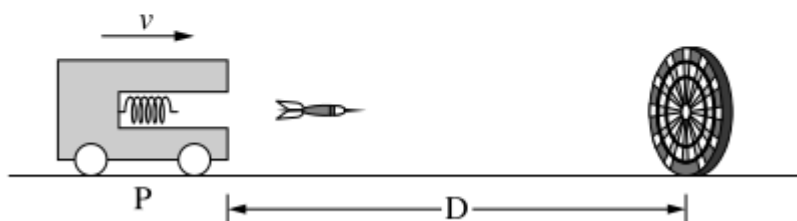
- (A) Projectile 1, because it will have a greater initial speed, which means it will have a larger average speed and therefore spend less time in the air.
 - (B) Projectile 2, because it will have a smaller initial vertical velocity.
 - (C) Projectiles 1 and 2 hit the ground at the same time because they have the same initial speed.
 - (D) The flight time cannot be compared without knowing the relative initial speeds.
- 139.** A rowboat moving in a lake is accelerating at a constant rate towards the shore. An observer in another boat moving at a constant velocity measures the row boat's acceleration. A second observer, standing still on the shore, also measures the acceleration of the rowboat. Are the two accelerations measured by the two observers the same? Why or why not?
- (A) No, because the observer on the other boat is in a non-inertial reference frame.
 - (B) No, because the observers have different velocities relative to the rowboat.
 - (C) Yes, because both observers are in inertial reference frames.
 - (D) Yes, because both observers are external to the rowboat's frame of reference and will therefore measure zero acceleration.

U1

140.



Trial 1



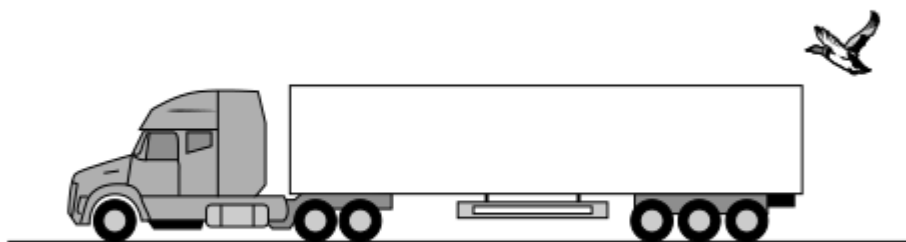
Trial 2

A cart equipped with a launcher launches a dart horizontally, as shown. The dart is always given the same horizontal velocity v_{dart} relative to the cart. In Trial 1, the cart is at rest at Point P a distance D from a target and launches the dart toward the target. The dart takes a time t_1 to reach the target. In Trial 2, the cart is moving toward the target with constant speed v_{cart} and launches the dart at the instant the cart is at Point P. The dart takes a time t_2 to reach the target. The cart has a negligible change in speed during the launch. Which of the following is a correct expression for the cart's speed v_{cart} ?

- (A) $\frac{D}{t_1 + t_2}$
- (B) $\frac{D}{t_1 - t_2}$
- (C) $\frac{D}{t_1} + \frac{D}{t_2}$
- (D) $\frac{D}{t_2} - \frac{D}{t_1}$

U1

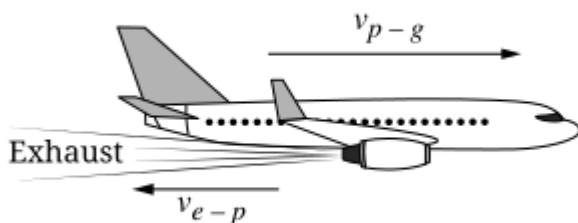
141.



A duck flies past a truck that is at rest. The duck takes 2.1 seconds to fly at constant speed v_d from the back of the truck to the front. Later, the truck travels along a flat, straight roadway with a constant speed $\frac{3}{4}v_d$. The duck again flies past the truck with constant speed v_d , flying in the same direction that the truck is moving. How long does it take the duck to fly from the back of the truck to the front in this case?

- (A) 1.2 seconds
- (B) 1.6 seconds
- (C) 2.8 seconds
- (D) 8.4 seconds

142.



A plane flies horizontally with constant speed v_{p-g} relative to the ground. While in flight, the plane ejects exhaust in the opposite direction at a speed v_{e-p} relative to the plane. Which expression correctly represents the velocity of the exhaust relative to the ground?

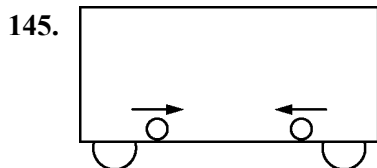
U1

- (A) $\sqrt{(v_{p-g})^2 + (v_{e-p})^2}$
 (B) $-v_{e-p}$
 (C) $v_{p-g} + v_{e-p}$
 (D) $v_{p-g} - v_{e-p}$

143. Three vehicles pass each other traveling on a three-lane roadway. Vehicle A is moving west with speed v_A relative to the road; Vehicle B is moving east with speed v_B relative to the road; and Vehicle C is moving east with speed v_C relative to the road. Assume east is the positive direction. What is the velocity of Vehicle B relative to observers in Vehicle A and in Vehicle C?

	Relative to Vehicle A	Relative to Vehicle C
A	$v_B + v_A$	$v_B + v_C$
B	$v_B + v_A$	$v_B - v_C$
C	$v_B - v_A$	$v_B + v_C$
D	$v_B - v_A$	$v_B - v_C$

- (A) A
 (B) B
 (C) C
 (D) D
144. A train is moving to the east at 25 m/s relative to the ground. A passenger at the back of the train throws a ball to the west with an initial speed of 15 m/s relative to the train. What is the initial velocity of the ball relative to the ground?
- (A) 10 m/s to the east
 (B) 10 m/s to the west
 (C) 40 m/s to the east
 (D) 40 m/s to the west



A train is traveling east with constant speed v_t . Two identical spheres are rolling on the floor of one train car. In the frame of reference of the train, the spheres are moving directly toward each other at one instant with the same speed v_p parallel to the train's motion, as shown in the figure above. What is the velocity of the center of mass of the spheres in the frame of reference of the train and in the frame of reference of a person standing at rest alongside the train?

U1

(A)	Train	Person
	Zero	Zero
(B)	Train	Person
	Zero	v_t east
(C)	Train	Person
	v_t east	$v_p + v_t$ east
(D)	Train	Person
	$v_p + v_t$ east	Zero

146. A student rolls a ball along the floor so that the ball has a speed of 16 m/s. What is the speed of the ball relative to a second student who is running at 6 m/s in the same direction that the ball is rolling?

- (A) 6 m/s
- (B) 10 m/s
- (C) 16 m/s
- (D) 22 m/s